

Enhancing Solubilization and Purification of M4: A Recombinant Membrane Protein Domain

*A Graduate Project Report submitted to Manipal Academy of Higher Education
in partial fulfilment of the requirement for the award of the degree of*

BACHELOR OF TECHNOLOGY
in

Biotechnology

Submitted by

Manthan S Joshi

Under the guidance of

Dr Vaishnavo Pai
Lead Scientist
Tranalab Pvt Ltd

Dr Subbalaxmi S
Assistant Professor - Selection Grade
MIT, Manipal



DEPARTMENT OF BIOTECHNOLOGY
MANIPAL INSTITUTE OF TECHNOLOGY
(A Constituent College of *Manipal Academy of Higher Education*)
MANIPAL – 576104, KARNATAKA, INDIA

January 2023 – June 2023



DEPARTMENT OF BIOTECHNOLOGY
MANIPAL INSTITUTE OF TECHNOLOGY
(A Constituent Institute of Manipal Academy of Higher Education)
MANIPAL – 576 104 (KARNATAKA), INDIA

Manipal
30 June 2023

CERTIFICATE

This is to certify that the project titled **Enhancing Solubilization and Purification of M4: A Recombinant Membrane Protein Domain** is a record of the bonafide work done by **MANTHAN S JOSHI** (*Reg. No. 190924042*) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (B.Tech) in **BIOTECHNOLOGY** of Manipal Institute of Technology Manipal, Karnataka, (A Constituent Institute of Manipal Academy of Higher Education), during the academic year 2019 - 2023.

Dr Subbalaxmi
Assistant Professor - Selection Grade, Biotech.
M.I.T., MANIPAL

Dr. Balaji S.
HOD, Biotech.
M.I.T., MANIPAL

TRANALAB PRIVATE LIMITED

205, ANANDANILAYA, N.SY 11/1,12/1, AECS LAYOUT, CHINNAPPANAHALLI Bangalore-560037
CIN: U74999KA2017PTC107803; GST: 29AAGCT5003P1ZX



Bangalore
30 June 2023

CERTIFICATE

This is to certify that the project entitled **Enhancing Solubilization and Purification of M4: A Recombinant Membrane Protein Domain** was carried out by **MANTHAN S JOSHI** (*Reg. No. 190924042*) at **Tranalab Pvt Ltd, Centre for Human Genetics, Bangalore** under my guidance during **January 2023 to June 2023**.

Vaishnavo Pai

Dr Vaishnavo Pai
Lead Scientist
Tranalab Pvt Ltd,
Bangalore



ACKNOWLEDGMENTS

I am grateful to Cdr. (Dr.) Anil Rana, the Director of MIT, Manipal, for providing the necessary infrastructure and resources for conducting this project. I also thank Dr. S. Balaji, the Head of the Department of Biotechnology, for his valuable guidance and support throughout the project. My heartfelt thanks go to Dr. Subbalaxmi S., my Faculty Advisor, for her continuous encouragement and insightful inputs. I am also grateful to Mrs. Archana M. Rao and all the faculty members for their assistance during the course of this project.

During my internship at Tranalab Pvt Ltd, Centre for Human Genetics, Bangalore, I extend my deepest appreciation to Dr. Nitin Deshpande, the CEO of the company, and Dr. Chandramowli Ganesh, the CSO, for providing me with the opportunity to work in their esteemed organization. I am indebted to Dr. Vaishnavo Pai, the Lead Scientist and Lab In charge, whose guidance, expertise, and support were instrumental in the successful execution of the project. I am especially and immensely grateful to Srushti Kore, Cheshta Chawra, and Vaishnavi S. for their immense help and for sharing their knowledge, enabling me to learn and grow as a researcher, particularly for this project. I would also like to acknowledge the contributions of Tanushree Y, Vaishnavi A, Luminaa A, Vani BS, Palash Ghosh, and Ragini M., who were invaluable members of the lab team and provided valuable assistance throughout my internship.

ABSTRACT

Protein solubilization and purification are essential steps in the field of biotechnology for obtaining highly pure and functional proteins. In this study, we aimed to optimize the solubilization and purification protocol for a specific protein, M4, using Ni-NTA chromatography. The objective of this project was to develop an efficient and reliable method for obtaining soluble and pure M4 protein for further downstream applications. The significance of this work lies in the potential applications of M4 in various fields such as therapeutics and industrial biotechnology.

The methodology involved several key steps. Firstly, competent cells of BL21 (DE3) strain were prepared, transformed with plasmid containing gene of interest and the cells were induced with IPTG for protein expression. The cells were then harvested and lysed and the M4 protein was solubilized using two different binding buffers: Condition 1 and Condition 2. The solubilized protein was subjected to Ni-NTA chromatography for purification. Additionally, we explored the use of sugar affinity chromatography for assessing the refolding of M4 protein.

The results demonstrated that solubilization with Condition 2, which included Triton X 100 and Tween 20, led to higher protein yield compared to Condition 1. This novel approach enhanced the solubility and recovery of M4 protein during purification. However, the sugar affinity chromatography did not yield satisfactory results despite variations in elution buffers. These findings highlight the limitation of sugar affinity chromatography for M4 protein purification.

In summary, the optimized solubilization and purification protocol using Condition 2 showed promising results for obtaining soluble and pure M4 protein. The incorporation of Triton X 100 and Tween 20 in the solubilization step significantly improved the protein yield. However, the inefficiency of sugar affinity chromatography suggests the need for alternative elution strategies for M4 protein. Future studies could explore other affinity chromatography techniques or consider modifying the M4 protein construct to improve its refolding and purification. This work contributes to the understanding of protein purification methods and provides insights for further studies on M4 protein applications.

LIST OF TABLES

Table No	Table Title	Page No
2.3.11.1	Summary of References on Ni-NTA Purification for Recombinant Protein Isolation	19
2.3.12.1	Summary of References on Recombinant Protein Expression in <i>Escherichia coli</i>	21
2.3.13.1	Summary of References on Recombinant Protein Expression in <i>Escherichia coli</i>	22
3.3.1	List of equipment used during this project	42

LIST OF FIGURES

Figure No	Figure Title	Page No
2.3.12.1	IPTG Induction Mechanism for Protein Expression in BL21 (DE3) Host Cell (Barakat et al., 2022)	20
2.3.14.1	SDS PAGE Setup	23
2.2.3.1	Secondary Inoculated Media	32
2.2.4.1	IPTG Induced Media	33
3.2.5.1	Sonication Set-up	34
3.2.6.1	Solubilization on Tube Rotator	35
3.2.7.1	Purification Set-up with Buffers and Test Sample	36
3.2.7.2	Ni-NTA Chromatography of M4	37
3.2.8.1	Dialysis Bag with Buffer	38
3.2.8.2	Dialysis of M4	39
3.3.1	An illustration of the steps involved in expression and purification of M4.	44
4.2.1	Expression of M4 using IPTG BL21 (DE3)	45
4.2.2	Solubilization of M4 under Conditions 1 and 2 of Solubilization (refer 3.2.6)	46
4.2.3	Ni-NTA Chromatography Purification of M4 under Conditions 1 and 2 of Solubilization (refer 3.2.6)	47
4.2.4	Sugar Binding Chromatography for M4 under Conditions 1, 2 and 3 (refer 3.2.9)	48

Contents		
		Page No
Cover Page		i
Certificate		ii
Certificate		iii
Acknowledgements		iv
Abstract		v
List of Tables		vi
List of Figures		vii
Contents		viii
Chapter 1	INTRODUCTION	1
1.2	Introduction to the Area of Work	1
1.1	Introduction	1
1.3	Brief Present-Day Scenario Regarding the Work Area	1
1.4	Motivation to Do the Project Work	2
1.5	Objective of the Work	2
1.6	Target Specifications	3
1.7	Project Work Schedule	3
1.8	Organization of the Project Report	4
Chapter 2	BACKGROUND THEORY and/or LITERATURE REVIEW	6
2.1	Introduction	6
2.2	Introduction to the Project Title	6
2.3	Literature Review	7
2.3.1	Recombinant Proteins	7
2.3.2	DNA Cloning Technology	8
2.3.3	Importance of Recombinant Proteins	9
2.3.4	Vectors: Expression Vectors, Plasmid Vectors, and PET Vectors	10
2.3.5	Expression Systems	10
2.3.6	Advantages and Disadvantages of <i>E. coli</i> as an Expression System	12
2.3.6.1	Advantages	12
2.3.6.2	Disadvantages	12
2.3.7	Recombinant Protein Purification	13

		2.3.8	Soluble and Insoluble Protein Purification	14
		2.3.9	Formation of Inclusion Bodies	16
		2.3.10	Applications of M4	17
		2.3.11	Ni-NTA Purification	18
		2.3.12	<i>E. coli</i> BL21(DE3)	20
		2.3.13	Use of Tween 20 and Triton X-100 in Solubilization Step	22
		2.3.14	SDS-PAGE for Analysis	23
		2.3.15	Nanodrop for Analysis	24
	2.4	Summarized outcome of the literature review		25
	2.5	Theoretical discussions		27
	2.6	General analysis		28
	2.7	Mathematical derivations		29
Chapter 3		METHODOLOGY		30
	3.1	Introduction		30
	3.2	Methodology		30
		3.2.1	Competent Cells	30
		3.2.2	Transformation	31
		3.2.3	Primary and Secondary Inoculum	32
		3.2.4	IPTG Induction	33
		3.2.5	Sonication	34
		3.2.6	Solubilization	35
		3.2.7	Ni-NTA Chromatography	36
		3.2.8	Dialysis	38
		3.2.9	Sugar Binding Affinity Chromatography	39
	3.3	Component specifications		40
Chapter 4		RESULTS AND DISCUSSION		45
	4.1	Introduction		45
	4.2	Result analysis		45
	4.3	Significance of the Results Obtained		48
	4.4	Deviation from Expected Results and Justification		49
Chapter 5		CONCLUSION AND FUTURE SCOPE		51
	5.1	Brief Summary of the Work		51
	5.2	Conclusions		51
	5.3	Future Scope of Work		52
		5.3.1	Exploring alternative chromatographic techniques	52
		5.3.2	Protein engineering and tag optimization	52

		5.3.3	Investigation of alternative affinity ligands	52
REFERENCES				53
PROJECT DETAILS				63
PLAGIARISM CHECK REPORT				64

CHAPTER 1

INTRODUCTION

1.1 Introduction

This chapter provides an overview of the research work conducted on the M4, a membrane protein domain with important cellular functions. The chapter begins with a brief outline of the topics that will be discussed, followed by an introduction to the general area of work. A snapshot of the present-day scenario in the field is presented, highlighting the motivation behind this project work. The shortcomings of previous work and references are further explored in this chapter, highlighting the importance of our work in the current context. The discussion also encompasses the uniqueness of the methodology to be adopted and the significance of the potential end result. Finally, the chapter concludes by presenting the objectives of the work, target specifications, project work schedule, and an overview of the organization of the project report.

1.2 Introduction to the Area of Work

The M4 protein is a membrane protein domain that plays a crucial role in various cellular processes, including signal transduction and cellular communication. Understanding the structure, function, and purification of M4 is of great importance in elucidating its biological functions and potential applications.

1.3 Brief Present-Day Scenario Regarding the Work Area

Currently, there is significant interest in studying membrane protein domains involved in cellular signaling pathways. The field of protein research has witnessed remarkable advancements, with a growing focus on understanding the molecular mechanisms underlying cellular functions. M4, being a key player in these processes, has gained considerable attention in recent years. However, despite the progress made in the field, there are still several challenges associated with the expression and purification of M4. Addressing these challenges is vital for furthering our understanding of M4 and its potential applications in various research areas.

1.4 Motivation to Do the Project Work

The motivation behind this project work stems from the limitations observed in previous studies and reference papers. The existing methodologies for expressing and purifying M4 have shown incomplete solubilization and low yield of the protein of interest. These shortcomings hinder the progress in studying the structure and function of M4 and limit its potential applications in research. Therefore, there is a need for an improved methodology that can enhance the solubilization and purification efficiency of M4.

The potential impact of advancing our understanding of M4 and its role in cellular signaling pathways is emphasized by the importance of this work in the present context. By overcoming the challenges associated with M4 expression and purification, we aim to contribute to the existing knowledge and pave the way for further investigations into the functional properties and potential applications of M4.

The methodology that we will adopt for solubilization protocol modification is unique. By incorporating additional components such as Tween 20 and Triton X-100, we observed improved solubilization of M4 compared to the conventional protocol. This modification sets our methodology apart and can potentially lead to higher yields of the desired protein.

The significance of the possible end result is multifaceted. First and foremost, successfully expressing and purifying M4 using the proposed methodology will provide researchers with a valuable tool to study its structure, function, and interactions. This knowledge can contribute to the development of targeted therapies and interventions for various cellular processes. Moreover, our work may serve as a foundation for future investigations into the potential role of M4 in different cellular signaling pathways.

1.5 Objective of the Work

1. Optimize the expression conditions for the M4 protein in *E. coli* BL21(DE3) to achieve high levels of protein production.
2. Design and implement a robust solubilization protocol for the M4 protein, aiming to maximize its solubility and recovery.
3. Perform Ni-NTA affinity chromatography to selectively bind the M4 protein to the resin, exploiting its polyhistidine (His) tag.
4. Develop a stepwise elution strategy for the M4 protein from the Ni-NTA resin, gradually decreasing the imidazole concentration to promote protein release.
5. Implement affinity chromatography using a specific resin with sugar-binding properties to further purify the M4 protein and remove contaminants.

6. Explore various elution conditions and buffers to effectively elute the tightly bound M4 protein from the sugar affinity chromatography resin, such as adjusting the pH, salt concentration, or introducing competitive ligands.
7. Evaluate the purity and quality of the eluted M4 protein using SDS-PAGE or other analytical techniques.
8. Conduct dialysis against an appropriate buffer to remove unwanted solutes and optimize the protein's buffer composition for downstream applications.
9. Obtain a purified and highly functional M4 protein suitable for further characterization, structural analysis, or functional studies.

1.6 Target Specifications

The end result of this project holds significant importance. Successfully expressing and purifying M4 using the proposed methodology will provide a pure and soluble form of the protein for further characterization and functional studies. This will enable researchers to investigate the structure, binding properties, and potential applications of M4. The target specifications include obtaining a sufficient yield of purified M4 with high purity and maintaining its structural and functional integrity.

1.7 Project Work Schedule

Month 1 (January):

- Familiarized myself with the Standard Operating Procedure (SOP) for the project.

Months 2 and 3 (February and March):

1. Spent these months familiarizing myself with the techniques required for the project.
2. Studied and understood the principles and protocols involved in each step of the project.

Months 4, 5, and 6 (April, May, and June):

- Undertook the project handling phase, which includes the following tasks:
 1. Competent Cells:
 - Prepared competent cells for efficient transformation.
 - Performed checks to ensure the competency of the cells.
 2. Transformation:
 - Performed the transformation step to introduce the desired DNA construct into the competent cells.
 3. Primary Inoculum:
 - Prepared a primary inoculum to initiate the growth of the transformed cells.
 4. Secondary Inoculum:

- Prepared a secondary inoculum by transferring cells from the primary inoculum to a larger culture volume.
5. IPTG Induction + SDS PAGE:
 - Induced protein expression using IPTG (Isopropyl β -D-1-thiogalactopyranoside).
 - Performed SDS-PAGE analysis to assess protein expression levels.
 6. Sonication:
 - Employed sonication to disrupt the cells and release the expressed proteins.
 7. Solubilization + SDS PAGE:
 - Solubilized the target protein obtained from the sonicated cells.
 - Verified the solubilization efficiency through SDS-PAGE analysis.
 8. Purification + SDS PAGE:
 - Employed the chosen purification technique (e.g., Ni-NTA chromatography) to purify the target protein.
 - Monitored the purification progress using SDS-PAGE analysis.
 9. Dialysis + SDS PAGE:
 - Performed dialysis to remove unwanted molecules and exchange the buffer for the purified protein.
 - Verified the effectiveness of dialysis through SDS-PAGE analysis.
 10. Sugar Affinity Chromatography + SDS PAGE:
 - Utilized sugar affinity chromatography to assess the successful refolding of the protein after dialysis.
 - Monitored the refolding efficiency using SDS-PAGE analysis.

1.8 Organization of the Project Report

Chapter 1	INTRODUCTION
1.1	Introduction
1.2	Introduction to the Area of Work
1.3	Brief Present-Day Scenario Regarding the Work Area
1.4	Motivation to Do the Project Work
1.5	Objective of the Work
1.6	Target Specifications
1.7	Project Work Schedule
1.8	Organization of the Project Report
Chapter 2	BACKGROUND THEORY and/or LITERATURE REVIEW
2.1	Introduction

2.2	Introduction to the Project Title
2.3	Literature Review
2.4	Summarized outcome of the literature review
2.5	Theoretical discussions
2.6	General analysis
2.7	Mathematical derivations
Chapter 3	METHODOLOGY
3.1	Introduction
3.2	Methodology
3.3	Component specifications
Chapter 4	RESULTS & DISCUSSION
4.1	Introduction
4.2	Result analysis
4.3	Significance of the Results Obtained
4.4	Deviation from Expected Results and Justification
Chapter 5	CONCLUSION AND FUTURE SCOPE
5.1	Brief Summary of the Work
5.2	Conclusions
5.3	Future Scope of Work

CHAPTER 2

BACKGROUND THEORY

2.1 Introduction

In this chapter, we delve into the background theory and literature review related to recombinant protein expression and purification. We explore various concepts and techniques that form the foundation of this field, providing a comprehensive understanding of the processes involved. The chapter begins with an overview of recombinant proteins, their importance, and the significance of their production through DNA cloning technology. We then discuss the role of vectors, expression systems, and the purification methods employed to obtain pure recombinant proteins. Additionally, we explore the formation of inclusion bodies, the use of solubilization agents such as Tween 20 and Triton X-100, and the analysis techniques of SDS-PAGE and Nanodrop. Throughout the chapter, we will specifically focus on the relevant aspects and techniques applicable to our project title, "Enhancing Solubilization and Purification of M4: A Recombinant Membrane Protein Domain." This chapter serves as the groundwork for understanding the subsequent experimental procedures and findings discussed in the later chapters of this thesis.

2.2 Introduction to the Project Title

The project title, "Enhancing Solubilization and Purification of M4: A Recombinant Membrane Protein Domain" specifically focuses on the expression and purification of the M4 protein using recombinant DNA technology. The M4 protein, being a membrane protein domain, plays a crucial role in cellular signaling pathways and is of significant interest in various fields, including biomedical research and therapeutics. In this chapter, we will explore the literature surrounding recombinant protein expression and purification, with a specific emphasis on the techniques and methodologies relevant to the successful expression and purification of the M4 protein. We will investigate the latest developments in the field, discuss the importance and applications of recombinant proteins, and review the different purification methods employed, including Ni-NTA chromatography. Furthermore, the use of *E. coli* BL21(DE3) as the expression host and the solubilization steps involving the use of Tween 20 and Triton X-100 will be examined. The chapter will also provide an overview of the analysis techniques utilized, such as SDS-PAGE and Nanodrop, to assess the purity and concentration of the expressed M4 protein. By delving into the background theory and literature surrounding these topics, we aim to establish a solid foundation for the subsequent experimental work and discussions presented in the following chapters.

2.3 Literature Review

In this sub-chapter, a comprehensive literature review will be presented, highlighting the present state and recent developments in the work area of recombinant protein expression and purification.

2.3.1 Recombinant Proteins

A class of proteins known as recombinant proteins is generated through the use of genetic engineering techniques, wherein the desired protein is encoded by specific DNA sequences introduced into host cells (Barken et al., 2007). Large quantities of the protein of interest can be produced using this process, offering advantages in terms of scalability and cost-effectiveness compared to traditional protein purification methods (Baneyx, 1999).

Recombinant proteins find importance in various fields, including biotechnology, medicine, and basic research. They have revolutionized the study of protein structure and function, enabling scientists to produce proteins in quantities sufficient for detailed characterization (Baneyx, 1999). They are used extensively in studying protein-protein interactions, as well as in the development of diagnostics, therapeutics, vaccines, and enzymes (Francis and Page, 2010). Recombinant proteins can also be modified with specific tags or labels, such as fluorescent proteins or affinity tags, which facilitate their purification and detection in complex biological samples (Francis and Page, 2010).

Vectors, which are DNA molecules, are used in the production of recombinant proteins to serve as carriers for the target DNA sequence. Plasmids, which are circular DNA molecules capable of replication within host cells, or other types of expression vectors specifically designed for protein expression, can be used as vectors (Baneyx, 1999). The choice of vector depends on various factors, including the host organism, desired protein yield, and the presence of specific regulatory elements required for efficient gene expression (Francis and Page, 2010).

Expression systems, such as bacteria (e.g., *E. coli*), yeast, insect cells, and mammalian cells, are used as hosts for the production of recombinant proteins. *E. coli* is one of the most widely used expression systems due to its ease of handling, rapid growth, and well-established genetic tools (Barken et al., 2007). It is particularly suitable for the expression of prokaryotic proteins and some eukaryotic proteins, including membrane proteins (Baneyx, 1999).

The purification of recombinant proteins is a critical step to obtain proteins of high purity and activity. Various purification techniques, such as affinity chromatography, size exclusion

chromatography, ion exchange chromatography, and hydrophobic interaction chromatography, are employed based on the physicochemical properties of the target protein (Baneyx, 1999). Affinity chromatography is commonly used for the purification of recombinant proteins, as it allows for the specific binding and elution of the protein of interest using affinity tags, such as histidine tags (His-tags) (Francis and Page, 2010).

2.3.2 DNA Cloning Technology

DNA cloning technology is a fundamental technique in molecular biology that enables the replication and manipulation of specific DNA sequences (Sambrook et al., 2001). It allows for the isolation and amplification of target DNA fragments, including those encoding recombinant proteins, for subsequent expression in host cells.

The cloning process involves the insertion of a DNA fragment of interest into a vector, typically a plasmid, to create a recombinant DNA molecule (Sambrook et al., 2001). Circular DNA molecules known as plasmids can replicate independently within host cells and carry genes conferring antibiotic resistance, which facilitates the selection and maintenance of transformed cells (Sambrook et al., 2001).

To clone a DNA fragment, several key steps are involved. First, the target DNA fragment and the vector are both cleaved with specific restriction enzymes, generating compatible ends that can be ligated together (Sambrook et al., 2001). Restriction enzymes recognize specific DNA sequences and cleave the DNA at or near these sites. The choice of restriction enzymes depends on the desired cloning strategy and the presence of suitable restriction sites in the DNA fragment and vector.

Following ligation, the recombinant DNA molecule is introduced into host cells through a process called transformation (Sambrook et al., 2001). The recombinant plasmids are taken up by host cells, such as *E. coli*, and are then replicated along with the host's own DNA during cell division.

Successful cloning is determined by selecting transformed cells that carry the recombinant plasmid. This selection is achieved using antibiotic resistance markers present on the plasmid (Sambrook et al., 2001). The transformed cells are grown on selective media containing the corresponding antibiotic, allowing only the transformed cells to survive and form colonies.

DNA cloning technology has greatly facilitated the production of recombinant proteins. It enables the isolation and amplification of specific DNA sequences, including those encoding target proteins, and their subsequent expression in host cells (Choi et al., 2004). The field of

molecular biology has been revolutionized by this technique, which provides researchers with a powerful tool for studying protein function, protein-protein interactions, and various biological processes.

2.3.3 Importance of Recombinant Proteins

Recombinant proteins have gained significant importance in various fields, including biotechnology, medicine, and basic research. Their versatility and potential applications have led to extensive research and development efforts aimed at producing these proteins for a wide range of purposes.

In the field of biotechnology, recombinant proteins are crucial for the production of enzymes used in industrial processes, such as the production of biofuels, pharmaceuticals, and agricultural products (Wurm et al., 2004). These proteins are also utilized in the development of diagnostics, allowing for the detection and monitoring of diseases, pathogens, and biomarkers (Raja and Natarajaseenivasan, 2015).

In medicine, recombinant proteins have revolutionized the development of therapeutic drugs. They serve as the active ingredients in a variety of biologics, including monoclonal antibodies, hormones, cytokines, and growth factors (Walsh, 2018). Recombinant proteins have been successfully employed in the treatment of various diseases, such as cancer, autoimmune disorders, and genetic disorders (Walsh, 2018; Glick and Patten 2022). They offer targeted therapy options with high specificity and reduced side effects compared to traditional small molecule drugs (Glick and Patten, 2022).

Furthermore, a crucial role is played by recombinant proteins in basic research, allowing protein structure, function, and interactions to be studied by scientists. They serve as valuable tools for investigating molecular mechanisms, cellular processes, and signaling pathways (Cedrone and Quéméneur, 2000). Recombinant proteins are utilized in assays, crystallography, spectroscopy, and other experimental techniques to gain insights into biological phenomena (Cedrone and Quéméneur, 2000).

The importance of recombinant proteins lies in their ability to provide a renewable and scalable source of proteins with defined properties and high purity (Walsh, 2018). Unlike native proteins isolated from natural sources, recombinant proteins can be produced in large quantities and modified to enhance their stability, solubility, and activity (Walsh, 2018). This allows for consistent and reliable production, making them ideal for research, diagnostics, and therapeutic applications.

2.3.4 Vectors: Expression Vectors, Plasmid Vectors, and PET Vectors

Vectors are essential tools in DNA cloning and recombinant protein expression. They are DNA molecules that are used to carry and replicate foreign DNA fragments, such as genes encoding proteins of interest, in host cells. Various types of vectors have been developed to accommodate different cloning strategies and host systems.

Researchers design expression vectors specifically for the efficient expression of cloned genes in host cells. Transcription and translation-facilitating elements, such as promoter regions, ribosome binding sites, and transcriptional terminators, are typically included in these vectors (Choi and Lee, 2004). Additionally, they often include selectable markers, such as antibiotic resistance genes, to enable the identification and selection of transformed cells.

Plasmid vectors are among the most commonly used types of vectors in molecular biology. They are small, circular DNA molecules that replicate independently within host cells. Plasmid vectors provide several advantages, including their stability, ease of manipulation, and ability to carry large DNA inserts (Shaner et al., 2005). These vectors are versatile and can accommodate various DNA fragments, making them suitable for different applications.

PET (polymerase chain reaction (PCR)-based expression) vectors are a specific type of plasmid vectors that have been optimized for high-throughput recombinant protein expression (Cormier et al., 2001). These vectors often contain multiple cloning sites for convenient insertion of DNA fragments, as well as affinity tags for purification purposes (Cormier et al., 2001). PET vectors are widely used in the production of recombinant proteins for structural and functional studies.

The choice of vector depends on several factors, including the host organism, the size of the DNA fragment to be cloned, the desired expression level, and the downstream applications. Different vectors have been developed for specific host systems, such as *E. coli*, yeast, insect cells, and mammalian cells, each with their own advantages and limitations (Choi and Lee, 2004). It is essential to carefully select the appropriate vector to ensure efficient gene expression and protein production.

2.3.5 Expression Systems

Expression systems play a critical role in the production of recombinant proteins. They are utilized to provide the necessary cellular machinery for the transcription and translation of cloned genes, resulting in the synthesis of the desired protein. Various expression systems have

been developed to accommodate different host organisms and specific requirements for protein production.

One commonly used expression system is the bacterial expression system, particularly *Escherichia coli* (*E. coli*). *E. coli* offers several advantages, including its well-characterized genetics, rapid growth rate, and cost-effectiveness (Mills et al., 2011). It has been extensively utilized for the production of recombinant proteins due to its ability to produce high yields of proteins in a relatively short time frame (Baneyx, 1999). *E. coli*-based expression systems are particularly suitable for the production of cytoplasmic and periplasmic proteins.

Another widely employed expression system is the yeast expression system, with *Saccharomyces cerevisiae* being the most commonly used yeast host. Yeast cells offer eukaryotic cellular machinery and can perform post-translational modifications, making them suitable for the production of eukaryotic proteins (Macauley-Patrick et al., 2005). Yeast expression systems have been successful in producing complex proteins, such as antibodies and enzymes, and are often employed in both academic and industrial settings (Macauley-Patrick et al., 2005; Ferndahl et al., 2010).

Insect cell-based expression systems, utilizing baculoviruses as vectors, are valuable tools for the production of complex and highly glycosylated proteins. The baculovirus expression vector system (BEVS) allows for the expression of recombinant proteins in insect cells, typically derived from *Spodoptera frugiperda* (Sf9) or *Trichoplusia ni* (High Five) (Unger and Peleg, 2012). BEVS has been utilized to produce a wide range of proteins, including viral antigens, enzymes, and therapeutic proteins (Unger and Peleg, 2012).

Additionally, mammalian cell expression systems have gained prominence for the production of complex proteins that require post-translational modifications and proper folding. Mammalian cells, such as Chinese hamster ovary (CHO) cells, offer the advantage of producing proteins with human-like glycosylation patterns and appropriate protein folding (Wurm, 2004). Mammalian expression systems are commonly used for the production of biopharmaceuticals, including monoclonal antibodies, hormones, and cytokines (Wurm, 2004; Stolt-Bergner et al., 2018).

The choice of an appropriate expression system depends on several factors, including the nature of the protein of interest, desired post-translational modifications, scalability, and downstream applications. Each expression system has its advantages and limitations, and the selection should be based on the specific requirements of the study or application.

2.3.6 Advantages and Disadvantages of *E. coli* as an Expression System

E. coli is one of the most commonly used expression systems for the production of recombinant proteins due to several advantages it offers. However, it is important to consider both the advantages and disadvantages when selecting *E. coli* as an expression host.

2.3.6.1 Advantages:

Well-established and cost-effective system: *E. coli* has been extensively studied and characterized, making it a well-established system for protein expression. The availability of a wide range of molecular tools, genetic manipulation techniques, and a large number of *E. coli* strains facilitate the production of recombinant proteins (Rosano and Ceccarelli, 2014).

High protein yields: *E. coli* can achieve high expression levels, allowing for the production of large quantities of recombinant proteins. The strong and tightly regulated promoters commonly used in *E. coli* expression systems, such as the T7 promoter, can drive high levels of protein synthesis (Rosano and Ceccarelli, 2014; Sørensen and Mortensen, 2005).

Fast growth and high cell density: *E. coli* has a rapid growth rate, allowing for quick production cycles. It can reach high cell densities in inexpensive culture media, leading to higher yields of recombinant proteins within a shorter time frame (Rosano and Ceccarelli, 2014; Sørensen and Mortensen, 2005).

Simplicity of downstream processing: *E. coli* offers ease of protein purification due to the availability of affinity tags, such as the polyhistidine (His-tag), that facilitate simple and efficient purification using immobilized metal affinity chromatography (IMAC) (Rosano and Ceccarelli, 2014; Terpe, 2003).

Despite its advantages, there are also certain limitations and challenges associated with using *E. coli* as an expression system.

2.3.6.2 Disadvantages:

Lack of post-translational modifications: *E. coli* is not capable of performing post-translational modifications (PTMs) commonly found in eukaryotic cells, such as glycosylation, phosphorylation, and disulfide bond formation. This limitation may affect the biological activity, stability, and immunogenicity of certain recombinant proteins (Rosano and Ceccarelli, 2014; Terpe, 2003).

Formation of inclusion bodies: Some recombinant proteins may have a tendency to form insoluble aggregates known as inclusion bodies when expressed in *E. coli*. This may occur due to incorrect folding, protein misfolding, or high protein expression levels. Refolding of inclusion bodies requires additional steps, such as solubilization and renaturation, which can be time-consuming and reduce protein yields (Rosano and Ceccarelli, 2014; Terpe, 2003).

Endotoxin contamination: *E. coli* is a gram-negative bacterium that naturally produces endotoxins, such as lipopolysaccharides (LPS). These endotoxins can contaminate recombinant protein preparations and may cause immune responses or interfere with downstream applications. Careful purification steps, such as endotoxin removal, are necessary to obtain endotoxin-free protein samples (Rosano and Ceccarelli, 2014; Terpe, 2003).

Limited capacity for complex protein folding: *E. coli* may face challenges in correctly folding and assembling complex proteins with multiple subunits or intricate disulfide bond formations. The lack of chaperones and other folding machinery present in eukaryotic systems may limit the expression of certain proteins in their native conformation (Rosano and Ceccarelli, 2014; Sørensen and Mortensen, 2005).

In conclusion, *E. coli* offers numerous advantages as an expression system, including cost-effectiveness, high yields, rapid growth, and simplified downstream processing. However, it is essential to consider the limitations associated with *E. coli*, such as the lack of PTMs, formation of inclusion bodies, endotoxin contamination, and limitations in folding complex proteins. These factors should be carefully evaluated when selecting *E. coli* as an expression host for recombinant protein production.

2.3.7 Recombinant Protein Purification

After successful expression of the target protein, the next crucial step is its purification. Purification is essential to isolate the recombinant protein from the complex cellular milieu and obtain a highly pure and bioactive form for further characterization and downstream applications. Various purification techniques and strategies have been developed to achieve efficient and specific separation of the target protein from contaminants.

One commonly used purification method is affinity chromatography, which exploits specific interactions between the target protein and a ligand immobilized on a solid support. A widely utilized affinity chromatography technique is the Ni-NTA (Nickel-nitrilotriacetic acid) chromatography, which utilizes the affinity of histidine residues present in recombinant proteins for metal ions, such as nickel (Fabian and Naumann, 2011). Ni-NTA chromatography

offers high selectivity and can efficiently capture and purify histidine-tagged proteins from complex mixtures (Terpe, 2003).

Another affinity chromatography technique often employed is the use of specific ligands, such as antibodies or receptors, immobilized on a solid support. This approach, known as immunoaffinity chromatography, enables the selective capture and purification of recombinant proteins based on their antigen-antibody interactions (Nelson and Cox, 2005). Immunoaffinity chromatography has been successfully applied to purify a wide range of proteins, including monoclonal antibodies and recombinant antigens (Nelson and Cox, 2005; Arnau et al., 2006).

In addition to affinity chromatography, other purification methods commonly employed include size exclusion chromatography, ion exchange chromatography, and hydrophobic interaction chromatography. Size exclusion chromatography separates proteins based on their size and molecular weight, allowing for the removal of larger aggregates and contaminants (Huse et al., 2002). Ion exchange chromatography utilizes the charge differences between the target protein and contaminants, enabling their separation based on their net charge (Liu et al., 2014). Hydrophobic interaction chromatography exploits the differences in hydrophobicity between the target protein and impurities, enabling their separation based on their affinity for hydrophobic ligands (Singh and Panda, 2005).

A comprehensive purification strategy often involves the combination of multiple purification techniques, known as a purification protocol. This strategy allows for the removal of different types of impurities and maximizes the purity and yield of the target protein. The choice and order of purification steps depend on the specific properties of the target protein, the presence of contaminants, and the desired purity level.

It is worth noting that the purification process can be challenging, particularly when dealing with insoluble or misfolded proteins. In such cases, strategies for solubilization, refolding, or denaturation followed by refolding may be required to obtain functional proteins (Berkmen, 2012). These additional steps aim to recover and properly fold the target protein from inclusion bodies or denatured states.

2.3.8 Soluble and Insoluble Protein Purification

During the process of recombinant protein expression, it is common to encounter both soluble and insoluble forms of the target protein. Soluble proteins are typically found in the cytoplasm or periplasmic space of the expression host, while insoluble proteins often form aggregates known as inclusion bodies (Villaverde and Mar Carrió, 2003). The purification of soluble and insoluble proteins requires different approaches and techniques.

Soluble protein purification involves the extraction and purification of the target protein from the soluble fraction of the cell lysate. The initial steps of soluble protein purification include cell lysis to release the protein and separation of the insoluble cell debris by centrifugation or filtration. Various chromatographic methods can be used to further purify the soluble protein of interest contained in the resulting supernatant (Sørensen et al., 2016). Soluble protein purification commonly employs chromatographic techniques such as affinity chromatography, ion exchange chromatography, and size exclusion chromatography (Shukla and Thömmes, 2010).

Affinity chromatography utilizes specific interactions between the target protein and an immobilized ligand, enabling the selective capture and purification of the protein. Common affinity chromatography ligands include metal ions, antibodies, and receptors (Hage, 2002). The choice of affinity ligand depends on the specific properties of the target protein and the desired purification strategy.

Ion exchange chromatography separates proteins based on their net charge. This technique utilizes charged resins with either a positively or negatively charged functional group to attract and bind oppositely charged proteins (Sørensen et al., 2016). By manipulating the buffer conditions, such as pH and ionic strength, the bound proteins can be eluted in a stepwise manner or with a gradient elution.

Size exclusion chromatography, also known as gel filtration chromatography, separates proteins based on their size and molecular weight. It relies on the differential diffusion rates of proteins through a porous gel matrix. Larger proteins elute earlier, while smaller proteins penetrate the gel matrix more deeply and elute later (Chico et al., 2013). Size exclusion chromatography is often used as a polishing step to further purify the target protein and remove remaining contaminants.

On the other hand, insoluble protein purification involves the solubilization and refolding or renaturation of the protein from inclusion bodies. Inclusion bodies are formed when the target protein aggregates in a misfolded or partially folded state within the host cells (Costa et al., 2014). To extract the insoluble protein, the inclusion bodies are typically isolated by cell lysis and subsequent centrifugation or filtration steps. The inclusion bodies can then be solubilized using denaturing agents such as urea or guanidine hydrochloride, followed by refolding through dialysis or rapid dilution methods (Costa et al., 2014).

The solubilization of inclusion bodies requires careful optimization of the solubilization buffer, denaturant concentration, and pH conditions. The denaturant disrupts the protein aggregates, allowing the protein to unfold and solubilize. Refolding involves the removal of the denaturant

and the creation of suitable conditions for proper protein folding (Costa et al., 2014). This process can be challenging and may require the addition of folding aids, such as chaperones or redox agents, to assist in the correct folding of the protein.

2.3.9 Formation of Inclusion Bodies

In the field of recombinant protein expression, the formation of inclusion bodies is a common phenomenon that occurs when the target protein accumulates in a misfolded or aggregated state within the host cells. Inclusion bodies are dense protein aggregates found in the cytoplasmic or periplasmic space, and they are composed of insoluble and biologically inactive protein particles (Carrió et al., 2005).

The formation of inclusion bodies can be attributed to various factors, including the overexpression of recombinant proteins, codon usage, protein folding kinetics, and host cell characteristics (Taglialegna et al., 2016). The accumulation of misfolded proteins overwhelms the cellular machinery responsible for correct protein folding and quality control, leading to the aggregation and sequestration of the protein into inclusion bodies (Carrió et al., 2005).

Despite their reputation as insoluble and non-functional protein aggregates, inclusion bodies have gained attention as potential sources of biologically active proteins. They offer several advantages in terms of protein stability, ease of purification, and potential for refolding (Carrió et al., 2005). Moreover, inclusion bodies can protect the host cells from toxic effects associated with the accumulation of misfolded proteins (Carrió and Villaverde, 2001).

To utilize inclusion bodies for protein production, the first step is the isolation and extraction of the inclusion bodies from the host cells (De Laureto et al., 2003). This can be achieved through cell disruption methods such as sonication or homogenization, followed by centrifugation or filtration to separate the insoluble inclusion bodies from the soluble cellular components (Carrió et al., 2005; Taglialegna et al., 2016).

Once isolated, the inclusion bodies are typically subjected to solubilization and refolding processes to recover the biologically active form of the target protein. Solubilization of inclusion bodies can be achieved using denaturing agents such as urea or guanidine hydrochloride, which disrupt the protein aggregates and unfold the target protein (Carrió and Villaverde, 2001). The solubilized proteins are then refolded by removing the denaturants and providing suitable refolding conditions, including controlled protein renaturation, redox buffers, chaperones, and specific folding agents (Carrió et al., 2005; Taglialegna et al., 2016).

It is important to note that the solubilization and refolding of inclusion bodies can be a challenging and time-consuming process, requiring optimization of various parameters such as pH, temperature, and protein concentration. The efficiency of refolding can vary depending on the protein of interest and the specific refolding strategy employed (Carrió et al., 2005; Carrió and Villaverde, 2001).

In recent years, strategies have been developed to improve the solubility and folding of proteins during expression, aiming to reduce the formation of inclusion bodies. These strategies include the use of fusion tags, co-expression of molecular chaperones, optimization of growth conditions, and modulation of protein expression levels (Taglialegna et al., 2016; Singh et al., 2015). These approaches aim to enhance the production of soluble and correctly folded proteins, reducing the need for inclusion body extraction and refolding procedures.

2.3.10 Applications of M4

The membrane protein domain (M4), as an important component of the immune system, has significant implications in biomedical research and therapeutic applications. Several studies have demonstrated the potential of M4 in various areas of research and therapy.

Biomedical Research: M4 has been extensively studied for its role in immune response modulation and inflammation regulation. Research has shown that M4 plays a crucial role in the recognition and clearance of pathogens, as well as the regulation of inflammatory responses (Bakeera-Kitaka et al., 2008; Zago et al., 2021). Understanding the mechanisms by which M4 functions can provide valuable insights into the development of novel therapies for infectious diseases, autoimmune disorders, and cancer.

Therapeutics: The unique properties of M4 make it a promising target for therapeutic interventions. Various approaches have been explored to harness the therapeutic potential of M4 in treating diseases. For example, researchers have shown promise in the treatment of autoimmune diseases such as rheumatoid arthritis and multiple sclerosis by developing monoclonal antibodies that target M4 (Pei and Yeo, 2016). Additionally, M4-based immunotherapies have demonstrated potential in cancer treatment by enhancing the immune response against tumor cells (Luo et al., 2018).

Drug Delivery Systems: M4 can also be utilized as a targeting moiety in drug delivery systems. By incorporating M4-specific ligands or peptides into nanocarriers, such as liposomes or nanoparticles, targeted delivery of therapeutic agents to M4-expressing cells can be achieved (Bakeera-Kitaka et al., 2008; Zago et al., 2021). This approach offers the advantage of improved drug efficacy and reduced off-target effects.

Biomaterials and Tissue Engineering: The use of M4 in biomaterials and tissue engineering has gained attention due to its role in cell adhesion and migration. Incorporating M4-derived peptides or mimetics into scaffolds or biomaterial surfaces can enhance cell-material interactions and promote tissue regeneration (Pei and Yeo, 2016; Luo et al., 2018). This application holds great potential for the development of functional biomaterials and tissue-engineered constructs.

Diagnostic Tools: M4 can also serve as a diagnostic biomarker in various diseases. Its expression levels or alterations in M4 function have been associated with certain pathological conditions, including inflammation-related disorders and cancer (Luo et al., 2018; Zago et al., 2021). Detection and monitoring of M4 expression can aid in disease diagnosis, prognosis, and therapeutic response evaluation.

2.3.11 Ni-NTA Purification

Ni-NTA (Nickel-nitrilotriacetic acid) affinity chromatography is a widely used technique for the purification of recombinant proteins, including the M4 protein. The principle behind Ni-NTA purification is the specific binding of polyhistidine (His-tagged) proteins to a Ni-NTA resin, allowing for the selective isolation of the target protein from complex mixtures.

The Ni-NTA purification process involves several steps to achieve high purity and yield of the M4 protein:

Binding: The first step is the binding of the His-tagged M4 protein to the Ni-NTA resin. The resin contains immobilized Ni²⁺ ions that interact with the exposed histidine residues on the protein of interest. This interaction is highly specific, allowing for the selective binding of the His-tagged M4 protein while other proteins in the mixture pass through.

Washing: After binding, the resin is washed with specific buffers to remove non-specifically bound impurities. The washing step is crucial for the removal of contaminants that might have weak interactions with the resin or non-specifically adhere to the His-tagged M4 protein. Multiple washes using buffers with decreasing concentrations of urea are commonly employed to achieve efficient purification.

Elution: Elution is the final step where the His-tagged M4 protein is released from the Ni-NTA resin. This is typically achieved by using a buffer with a high concentration of imidazole, which competes with the histidine residues on the protein for binding to the Ni²⁺ ions. Elution

conditions need to be optimized to obtain a high yield of purified M4 protein while maintaining its stability and functionality.

Several studies have utilized Ni-NTA purification for the isolation of various recombinant proteins, showcasing its effectiveness and versatility:

Table 2.3.11.1 Summary of References on Ni-NTA Purification for Recombinant Protein Isolation

Author (Year)	Description
Hochuli et al., 1987	A new metal chelate adsorbent selective for proteins and peptides containing neighboring histidine residues has been developed.
Kapust et al., 2002	Investigated the P1' specificity of tobacco etch virus protease.
Cheung et al., 2012	Expressed and purified histidine-tagged recombinant human prolactin using immobilized metal affinity chromatography.
Schmidt and Skerra, 1994	Proposed a one-step affinity purification method using the "Strep tag" and immobilized recombinant core streptavidin.
Terpe, 2003	Provided an overview of tag protein fusions, covering molecular and biochemical fundamentals and commercial systems.

The Ni-NTA purification technique provides a powerful tool for the purification of the M4 protein, enabling the isolation of highly pure and functional protein for downstream applications. Optimization of the binding, washing, and elution conditions is essential to achieve the desired level of purity and yield.

2.3.12 *E. coli* BL21(DE3)

E. coli BL21(DE3) is a commonly used bacterial strain for the expression of recombinant proteins, including the M4 protein. It is derived from *E. coli* B, a non-pathogenic strain, and carries a mutated *lacI* gene and a chromosomal copy of the T7 RNA polymerase gene (DE3). These genetic modifications enable the tightly regulated expression of target genes under the control of the T7 promoter upon induction with IPTG (isopropyl β -D-1-thiogalactopyranoside).

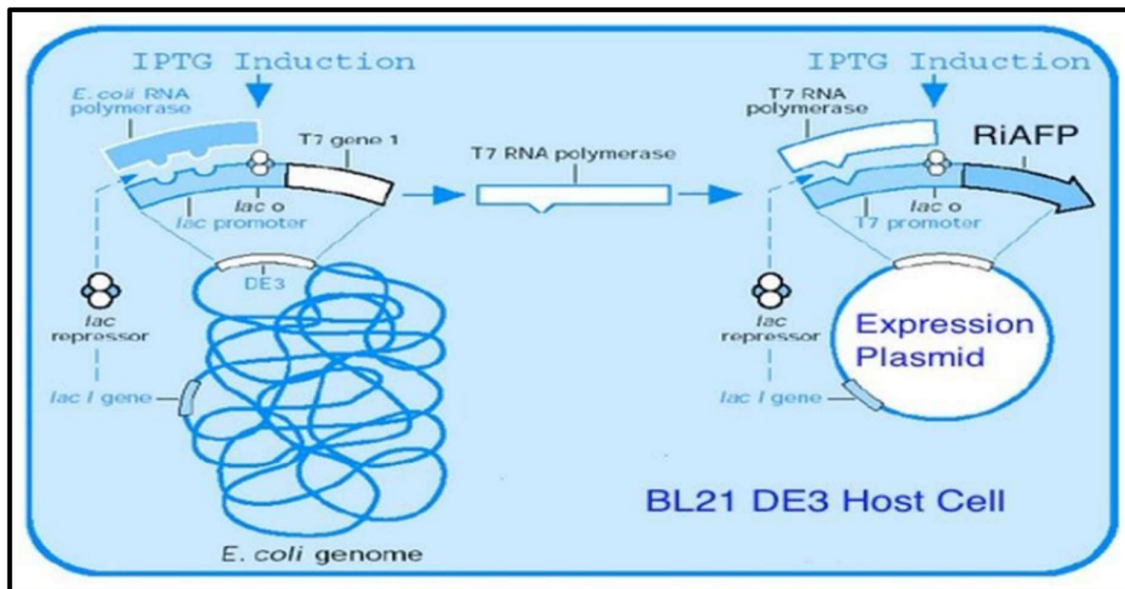


Figure 2.3.12.1 IPTG Induction Mechanism for Protein Expression in BL21 (DE3) Host Cell (Barakat et al., 2022)

E. coli BL21(DE3) offers several advantages for recombinant protein expression:

High expression levels: The presence of the T7 RNA polymerase allows for efficient transcription from the T7 promoter, leading to high levels of target protein production. This makes *E. coli* BL21(DE3) suitable for the expression of M4 protein in large quantities.

Compatibility with plasmid vectors: *E. coli* BL21(DE3) is compatible with a wide range of plasmid vectors commonly used for recombinant protein expression, such as pET vectors. These vectors provide the necessary elements for gene cloning, expression, and selection in bacteria.

Genetic stability: *E. coli* BL21(DE3) exhibits good genetic stability, ensuring the consistent expression of the target protein over multiple generations. This stability is important for maintaining the integrity of the M4 protein during large-scale production.

Several studies have utilized *E. coli* BL21(DE3) for the expression of various recombinant proteins, demonstrating its reliability and versatility:

Table 2.3.12.1 Summary of References on Recombinant Protein Expression in *Escherichia coli*

Author (Year)	Description
Studier et al., 1990	Described the use of T7 RNA polymerase for directing expression of cloned genes.
Esposito and Chatterjee, 2006	Explored the enhancement of soluble protein expression through the use of fusion tags.
Carpousis, 2007	The RNA degradosome of <i>Escherichia coli</i> , an mRNA-degrading machine, was discussed. It is assembled on RNase E.
Singha et al., 2017	Reviewed the progress and challenges in expressing recombinant proteins in <i>Escherichia coli</i> .
Hussain, 1984	Highlighted recent advances in inducible protein expression systems in <i>Escherichia coli</i> .

The use of *E. coli* BL21(DE3) as an expression host for the M4 protein provides a robust platform for efficient and scalable production. The combination of its high expression levels, compatibility with plasmid vectors, and genetic stability make it a popular choice for recombinant protein expression in research and industrial settings.

2.3.13 Use of Tween 20 and Triton X-100 in Solubilization Step

The solubilization step plays a crucial role in the purification of recombinant proteins, including the M4 protein, particularly when dealing with insoluble protein aggregates or inclusion bodies. To improve solubilization efficiency, detergents such as Tween 20 and Triton X-100 are commonly employed.

Tween 20 (polyoxyethylene sorbitan monolaurate) and Triton X-100 (polyethylene glycol tert-octylphenyl ether) are non-ionic surfactants that aid in the solubilization and extraction of hydrophobic proteins. They act by disrupting hydrophobic interactions and reducing protein aggregation, thus facilitating the release of the target protein from inclusion bodies or insoluble fractions.

Several studies have demonstrated the effectiveness of Tween 20 and Triton X-100 in improving solubilization during protein purification:

Table 2.3.13.1 Summary of References on Recombinant Protein Expression in *Escherichia coli*

Author (Year)	Description
Okuda and Tokuda, 2009	Developed an <i>Escherichia coli</i> soluble protein releasing system for overproducing periplasmic proteins.
Ma et al., 2006	Overexpressed, purified, and functionally characterized a novel lipase from <i>Bacillus safensis</i> .
Chen et al., 2021	Produced soluble and active recombinant human thrombopoietin in <i>Escherichia coli</i> by fusing with SUMO.
Park et al., 2009	Expressed, purified, and characterized an antimicrobial peptide, human lactoferricin derived peptide, in <i>Escherichia coli</i> .

Malygin et al., 2003	Expressed, purified, and refolded recombinant human ribosomal protein S5 in <i>Escherichia coli</i> .
-------------------------	---

The addition of Tween 20 and Triton X-100 to the solubilization buffer improves protein solubility by disrupting hydrophobic interactions and reducing aggregation. However, the optimal concentration of these detergents may vary depending on the specific protein of interest and experimental conditions. Careful optimization is necessary to achieve efficient solubilization while maintaining protein integrity and functionality.

2.3.14 SDS-PAGE for Analysis

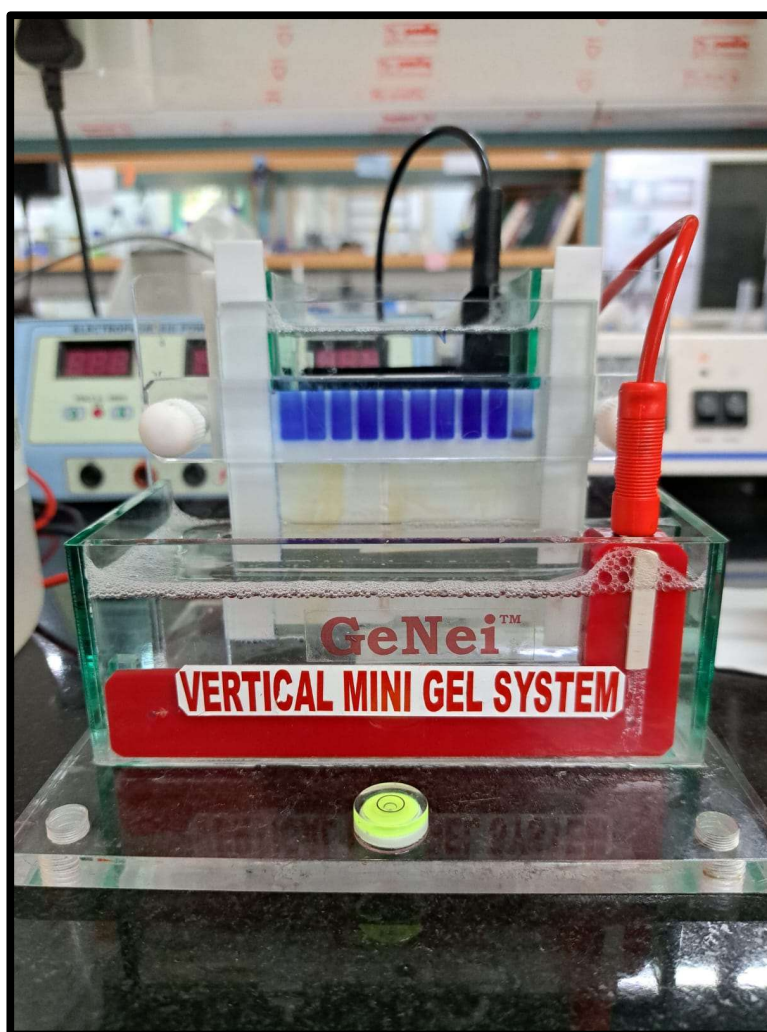


Figure 2.3.14.1 SDS PAGE Setup

SDS-PAGE (Sodium Dodecyl Sulfate-Polyacrylamide Gel Electrophoresis) is a widely used technique for the analysis of recombinant proteins. This method separates proteins based on their size and provides valuable information about the purity, molecular weight, and integrity of the target protein (Nice, 2021). In SDS-PAGE, proteins are denatured and coated with SDS, a detergent that imparts a negative charge to the proteins and unfolds them into linear polypeptide chains. The proteins are then separated in a polyacrylamide gel matrix based on their size, with smaller proteins migrating faster than larger ones (Laemmli, 1970).

The resolved protein bands in the gel can be visualized using various staining methods. Coomassie Brilliant Blue staining is commonly employed as it allows for the detection of protein bands in the nanogram to microgram range (Neuhoff et al., 1988). This staining method binds to proteins, producing a blue color, and provides a qualitative assessment of protein purity and the presence of contaminants or additional bands.

In addition to Coomassie staining, other staining methods such as silver staining or fluorescent dyes can be utilized for increased sensitivity or specific applications (Shevchenko et al., 1996; Candiano et al., 2004). Silver staining offers higher sensitivity and can detect proteins in the sub-nanogram range. It provides sharp bands with low background noise, making it useful for analyzing low-abundance proteins. Fluorescent dyes, such as SYPRO Ruby or Deep Purple, offer the advantage of quantitative analysis, as the dye binds stoichiometrically to proteins and can be detected using fluorescent scanners.

Furthermore, the size of the target protein can be estimated by comparing the migration pattern of the recombinant protein on the gel to protein markers of known molecular weight. This provides an important confirmation of the expected molecular weight and can aid in verifying successful protein expression and purification (Walker, 2005).

In summary, SDS-PAGE is a fundamental technique for analyzing recombinant proteins. It enables the assessment of protein purity, molecular weight determination, and the detection of contaminants or additional bands. The choice of staining method depends on the desired sensitivity and application, with Coomassie staining being the most commonly used method in routine protein analysis.

2.3.15 Nanodrop for Analysis

Nanodrop spectrophotometry is a widely used technique for the quantification and characterization of recombinant proteins. It offers a quick and convenient method for measuring the concentration and purity of proteins based on their UV-Vis absorbance spectra. The principle behind Nanodrop is the measurement of light absorption by the protein sample

at specific wavelengths, allowing for the determination of protein concentration and assessing the presence of contaminants (Vardakou et al., 2014).

The Nanodrop instrument utilizes a small sample volume (typically 1-2 μL), making it suitable for analyzing limited protein samples, such as those obtained from recombinant protein expression and purification processes. The instrument measures the absorbance of the protein sample at wavelengths of interest, such as 280 nm for assessing protein concentration and purity. The absorbance at 280 nm corresponds to the presence of aromatic amino acids, particularly tryptophan and tyrosine, which are abundant in proteins (Rosenblum et al., 1988).

Protein quantification using Nanodrop relies on the Beer-Lambert law, which states that the absorbance of a sample is directly proportional to its concentration. The protein concentration in the sample can be determined by measuring the absorbance at 280 nm and applying a specific extinction coefficient or using precalibrated methods (Schuler, 2005).

In addition to protein quantification, Nanodrop can also provide insights into protein purity. The presence of contaminants, such as nucleic acids or other impurities, can be detected by measuring the absorbance at different wavelengths. For example, the absorbance ratio at 260 nm to 280 nm (A_{260}/A_{280}) can be used to assess the purity of proteins by indicating the presence of nucleic acids, with a higher ratio suggesting contamination (Schuler, 2005).

Moreover, Nanodrop allows for the determination of protein secondary structure using the measurement of absorbance at specific wavelengths, such as 208 nm and 222 nm. These wavelengths correspond to the characteristic absorption peaks of α -helices and β -sheets, respectively. Changes in the secondary structure of recombinant proteins can provide valuable insights into their stability and conformational changes under different conditions (Cruz et al., 2010).

2.4 Summarized outcome of the literature review

The literature review conducted in this study provides valuable insights into the current state and recent developments in the field of recombinant protein expression. The following key outcomes emerged from the review:

Recombinant proteins: Recombinant proteins are artificial proteins generated by the expression of cloned genes in a host organism. They offer several advantages, such as scalability, cost-effectiveness, and the ability to produce proteins that are difficult to obtain from natural sources (Studier et al., 1990; Esposito and Chatterjee, 2006).

DNA cloning technology: DNA cloning technology plays a crucial role in recombinant protein expression. It involves the insertion of a target gene into a vector, such as a plasmid, to facilitate its replication and expression in a host organism (Studier et al., 1990; Carpousis, 2007).

Importance of recombinant proteins: Recombinant proteins have gained immense importance in various fields, including biomedical research and therapeutics. They serve as valuable tools for understanding protein structure and function, drug discovery, diagnostics, and the development of biopharmaceuticals (Singha et al., 2017; Hussain, 1984).

Vectors: Expression vectors, such as plasmid vectors and PET vectors, are commonly used in recombinant protein expression. These vectors contain essential elements, including promoter regions, ribosome binding sites, and selection markers, to ensure efficient gene expression (Krittanaï et al., 2018; Malygin et al., 2003).

Expression systems: Various expression systems, such as bacterial, yeast, insect, and mammalian systems, are available for recombinant protein production. Among them, *E. coli* is widely utilized due to its advantages, including high protein yields, cost-effectiveness, and well-established molecular tools (Singha et al., 2017; Hussain, 1984).

Protein purification: Purification of recombinant proteins is a critical step to obtain highly pure and functional proteins. Different purification methods, including affinity chromatography, size exclusion chromatography, and ion exchange chromatography, are employed to separate the target protein from impurities (Schmidt and Skerra, 1994; Chen et al., 2021).

Soluble and insoluble protein purification: Some recombinant proteins may form inclusion bodies, which are insoluble aggregates. Special strategies and techniques, such as solubilization and refolding, are required to recover the target protein from inclusion bodies (Ma et al., 2009).

Applications of recombinant proteins: Recombinant proteins find applications in various fields, including biomedical research, diagnostics, enzyme production, and the development of therapeutics. They are utilized in studying disease mechanisms, vaccine development, and as therapeutic agents themselves (Krittanaï et al., 2018; Chen et al., 2021).

These outcomes provide a comprehensive understanding of the current advancements, challenges, and potential applications of recombinant protein expression. They form the foundation for the methodology and experimental design in this research study.

2.5 Theoretical discussions

In this section, we will delve into the theoretical aspects and concepts relevant to recombinant protein expression. Theoretical discussions provide a solid foundation for understanding the underlying principles and mechanisms involved in the process. The following topics will be covered:

Gene Expression: DNA encodes genetic information that is transcribed into messenger RNA (mRNA) and subsequently translated into proteins, making gene expression a fundamental process. The regulation of gene expression is essential for controlling protein production (Rosano and Ceccarelli, 2014).

Transcription and Translation: Transcription is the process by which RNA polymerase synthesizes an mRNA molecule using a DNA template. Translation is the subsequent process where the mRNA is used as a template to synthesize a specific protein sequence by ribosomes (Rosano and Ceccarelli, 2014; Sørensen and Mortensen, 2005).

Codon Usage Bias: Codon usage bias refers to the unequal usage of synonymous codons encoding the same amino acid within an organism or between different organisms. It plays a role in the efficiency of translation and protein expression in recombinant systems (Sørensen and Mortensen, 2005; Malygin et al., 2003).

Factors Affecting Protein Expression: Several factors influence protein expression levels, including mRNA stability, promoter strength, codon usage, mRNA secondary structure, and translational efficiency. Optimizing these factors can enhance protein production (Rosano & Ceccarelli, 2014; Ma et al., 2006).

Protein Folding and Quality Control: Proper protein folding is crucial for functional protein production. Molecular chaperones and folding factors assist in the correct folding of recombinant proteins, while quality control mechanisms ensure the elimination of misfolded or aggregated proteins (Rosano and Ceccarelli, 2014; Cao et al., 2018).

Post-translational Modifications: Post-translational modifications (PTMs) play a vital role in protein function and stability. Common PTMs include phosphorylation, glycosylation, acetylation, and proteolytic cleavage. Understanding and controlling PTMs are important for producing functional recombinant proteins (Cao et al., 2018; Park et al., 2009).

Yield Optimization Strategies: To maximize protein yield, various strategies can be employed, including optimization of growth conditions, induction strategies, media composition, and co-

expression of chaperones or folding factors. These strategies aim to enhance protein production and improve the solubility of recombinant proteins (Hussain, 1984; Singha et al., 2017).

2.6 General analysis

In this section, we will provide a general analysis of the literature reviewed in the previous section. The analysis encompasses various aspects related to recombinant protein expression, including the choice of expression systems, protein purification techniques, and the challenges associated with optimizing protein yield and quality.

One of the key findings from the literature is the widespread use of *Escherichia coli* (*E. coli*) as a preferred host organism for recombinant protein expression (Studier et al., 1990; Ma et al., 2006; Singha et al., 2017). *E. coli* offers several advantages such as rapid growth, ease of genetic manipulation, high expression levels, and cost-effectiveness (Esposito and Chatterjee, 2006; Hussain, 1984). However, it is important to note that *E. coli* may not be suitable for the expression of all proteins, especially those requiring post-translational modifications or protein folding in more complex cellular environments (Singha et al., 2017; Cao et al., 2018).

To achieve efficient protein expression, various factors must be considered. These include the choice of expression vectors, promoters, and fusion tags. The selection of suitable expression vectors, such as plasmid vectors or PET vectors, depends on the specific requirements of the protein of interest (Terpe, 2003; Kapust et al., 2002). Similarly, the choice of promoters and fusion tags can significantly impact protein expression levels and solubility (Esposito & Chatterjee, 2006; Terpe, 2003).

Another crucial aspect of recombinant protein expression is protein purification. Several techniques have been employed for protein purification, including affinity chromatography, size exclusion chromatography, and ion exchange chromatography (Schmidt and Skerra, 1994; Krittanai et al., 2018). Among these, the Ni-NTA purification system has gained popularity due to its high selectivity and affinity for histidine-tagged proteins (Hochuli et al., 1987). Ni-NTA purification has been successfully used for the purification of various recombinant proteins, including enzymes, hormones, and antimicrobial peptides (Krittanai et al., 2018; Park et al., 2009).

During the literature analysis, it became evident that protein solubility is a significant challenge in recombinant protein expression. Many proteins tend to form insoluble aggregates known as inclusion bodies, leading to decreased yields and difficulties in downstream processing (Hussain, 1984). Strategies to enhance protein solubility include the use of chaperones,

optimization of growth conditions, and the addition of solubilizing agents such as tween 20 and triton X-100 during the solubilization step (Hussain, 1984; Malygin et al., 2003).

Furthermore, characterization and analysis of recombinant proteins are crucial to assess their quality and functionality. Techniques such as SDS-PAGE (sodium dodecyl sulfate-polyacrylamide gel electrophoresis) and Nanodrop spectroscopy have been widely employed for protein analysis (Ma et al., 2006; Park., 2009).

2.7 Mathematical derivations

In this section, we will present the mathematical derivations used for the calculation of protein concentration. The formula commonly employed is as follows:

$$C = A / (\epsilon \times l)$$

Where:

C: Protein Concentration ($\mu\text{g/ml}$)

A: Absorbance at 280 nm

ϵ : Extinction coefficient at 280 nm ($95965 \text{ M}^{-1}\text{cm}^{-1}$)

l: Path length in cm (10 mm)

This formula allows us to determine the protein concentration based on the absorbance measurement at 280 nm, taking into account the extinction coefficient and the path length. The extinction coefficient at 280 nm for M4 protein is typically considered as $95965 \text{ M}^{-1}\text{cm}^{-1}$ (Gill and von Hippel, 1989; Pace et al., 1995).

To obtain the final protein concentration in mg/ml , the calculated concentration is then multiplied by the molecular weight of the protein of interest. For example, if the protein has a molecular weight of 35 kDa, the final concentration would be determined by multiplying the calculated concentration by 35.

This mathematical derivation provides a straightforward and widely used method for determining protein concentrations based on absorbance measurements at 280 nm. It is important to note that the accuracy of the calculation relies on the appropriate selection of the extinction coefficient and the path length, as well as the assumption of a single dominant chromophore contributing to the absorbance at 280 nm.

CHAPTER 3

METHODOLOGY

3.1 Introduction

In this chapter, the methodology employed for the experimental procedures and data analysis will be presented. The primary objective of this chapter is to provide a detailed description of the methods and techniques used to address the research questions and achieve the objectives of the study. The chapter will outline the step-by-step procedures, equipment, materials, and protocols utilized in the research. Additionally, it will discuss the rationale behind the selection of specific methodologies and any modifications or adaptations made to existing techniques. By presenting a comprehensive overview of the methodology, this chapter aims to ensure the transparency and reproducibility of the study's findings and provide a foundation for the subsequent results and discussion.

3.2 Methodology

3.2.1 *Competent Cells*

The BL21 (DE3) cells were made competent by subjecting them to a heat shock treatment after resuspending them in ice-cold 10% glycerol. This process enhances their ability to take up and express foreign DNA efficiently (Joly et al., 1998).

In this study, a single colony of freshly revived BL21 (DE3) *E. coli* was inoculated in 3 mL of LB broth and incubated overnight at 37°C. From this culture, a 100 µL aliquot was used to inoculate a 50 mL LB broth, allowing it to reach an OD600 of 0.6 at 37°C. The culture was then transferred to a sterile tube and chilled on ice for 30 minutes. Subsequent centrifugation steps were performed at $4,500 \times g$ for 10 minutes at 4°C, followed by resuspension in ice-cold 100 mM CaCl₂. The cells were incubated on ice, centrifuged again, and resuspended in fresh ice-cold CaCl₂. After a final centrifugation step, the pellet was resuspended in 2 mL of CaCl₂ and stored at -80°C. These steps ensured the proper preparation of the *E. coli* culture for further experiments, providing an adequate supply of cells for subsequent analyses.

3.2.2 *Transformation*

The plasmid DNA was introduced into the competent BL21 (DE3) cells through a heat shock treatment, allowing the cells to uptake and express the desired genetic material. This step enables the incorporation of the target gene into the bacterial genome (Hanahan, 1983).

The competent cells were prepared as described previously by growing the cells, washing them with ice-cold sterile water, and resuspending them in 10% glycerol. The cells were then added with a certain amount of plasmid DNA, and the mixture was incubated on ice for 30 minutes. A heat shock treatment was applied to the cells at 42°C for 45 seconds, followed by immediate cooling on ice for 2 minutes. The cells were allowed to recover in SOC medium at 37°C for 1 hour with shaking before being plated on LB agar containing kanamycin. The successful transformation was confirmed by observing the growth of kanamycin-resistant colonies on the selective media.

3.2.3 Primary and Secondary Inoculum



Figure 2.2.3.1 Secondary Inoculated Media

The primary and secondary inoculum steps involve the propagation of the transformed BL21 (DE3) cells in a selective growth medium. The primary inoculum allows for the initial expansion of the transformed cells, while the secondary inoculum further amplifies the cell population for subsequent protein expression (Studier and Moffatt, 1986).

A small amount of the kanamycin resistant BL21 (DE3) strain was inoculated into 5 mL of LB broth containing kanamycin. To obtain a primary inoculum, the culture was incubated overnight at 37°C with shaking. The next day, a small volume of the primary inoculum was transferred to 50 mL of fresh LB broth containing kanamycin to obtain a secondary inoculum. The culture was incubated until it reached the desired optical density at 600 nm by maintaining a temperature of 37°C with continuous shaking.

3.2.4 IPTG Induction

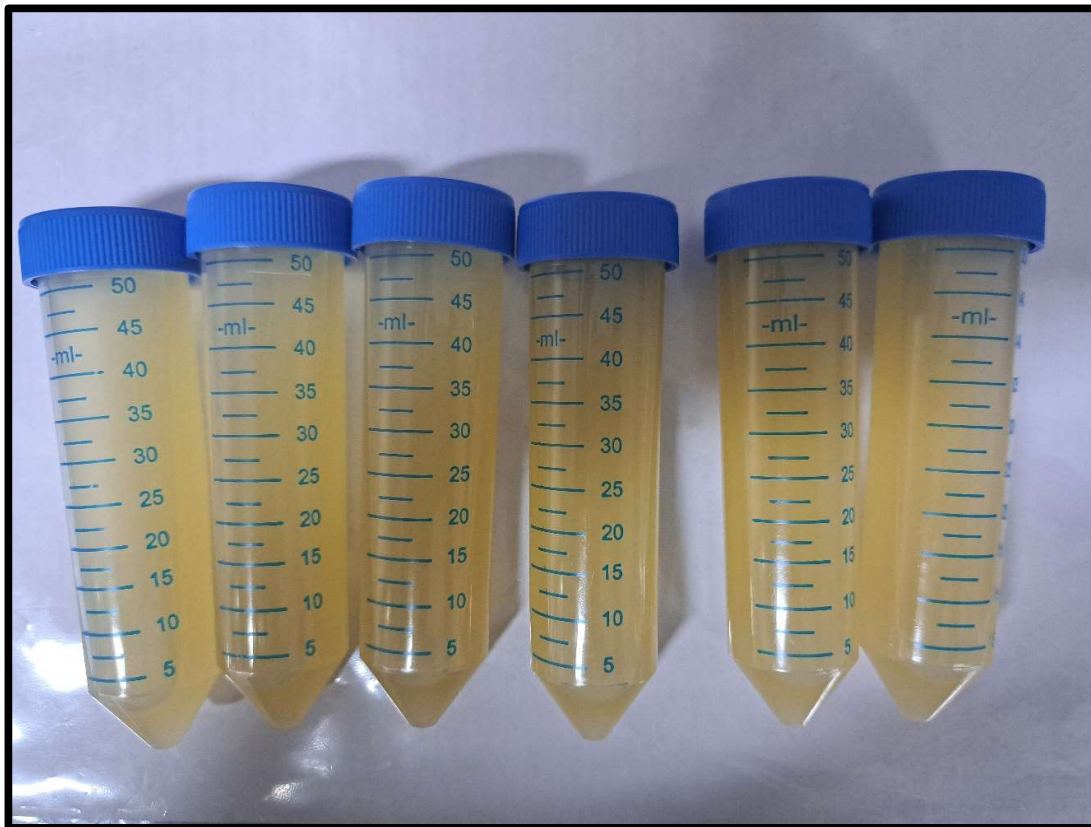


Figure 2.2.4.1 IPTG Induced Media

The addition of IPTG to the secondary inoculum induces the expression of the target protein by activating the lac operon in BL21 (DE3) cells. IPTG acts as a molecular mimic of lactose, thereby inducing the transcription of the gene of interest (Studier and Moffatt, 1986).

To induce protein expression in the secondary inoculum of kanamycin-resistant BL21 (DE3) cells, IPTG (isopropyl β -D-1-thiogalactopyranoside) was added to the culture at a final concentration of 1 mM. To allow for the expression of the target protein, the culture was incubated with shaking at 37°C for a certain period of time, typically 4-6 hours. The optimal induction time and IPTG concentration may vary depending on the specific protein and experimental conditions. The cells were subjected to centrifugation to harvest them after induction. The resulting cell pellet obtained could be either used immediately for downstream applications or stored at -80°C for future use. The success of protein expression was confirmed by analyzing the cell lysate via techniques such as SDS-PAGE and Coomassie staining.

3.2.5 Sonication

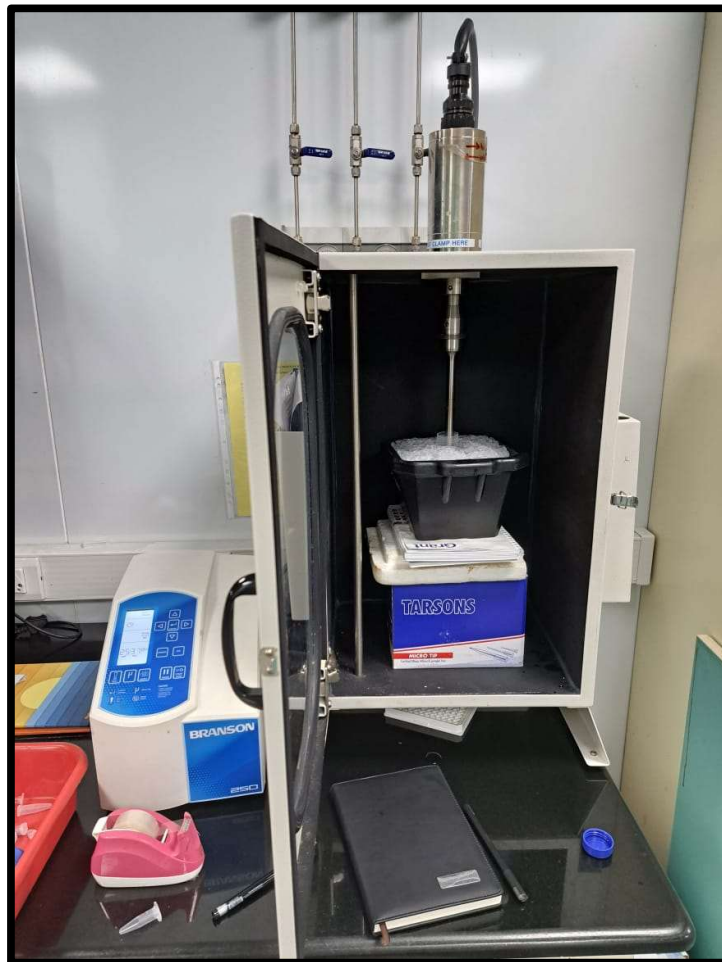


Figure 3.2.5.1 Sonication Set-up

Sonication is employed to disrupt the cells and release the intracellular proteins. The application of high-frequency sound waves causes the formation and collapse of bubbles, resulting in the mechanical disruption of the cell membranes and the liberation of the target protein into the surrounding buffer (Bushell et al., 2013).

To extract proteins from the above cell pellet, the cells were resuspended in a resuspension buffer containing 50 mM Tris base and 20 mM NaCl. The cell suspension was then subjected to sonication using a probe sonicator to break open the cells and release the proteins. The sonication was carried out on the ice for a total of 30 to 45 minutes, with short pulses of sonication followed by cooling on ice to prevent overheating of the sample. The optimal sonication time and intensity were: On time- 15 sec, Total on time- 15 mins, %A= 60%. After sonication, the cell debris was removed by centrifugation, and the resulting supernatant was collected for downstream applications such as protein purification or further analysis. The

protein concentration in the supernatant was determined using a suitable method such as Bradford assay or UV absorbance spectroscopy.

3.2.6 Solubilization



Figure 3.2.6.1 Solubilization on Tube Rotator

The solubilization step involves the resuspension of the protein pellet in a binding buffer containing urea and imidazole. This process allows for the complete dissolution of the target protein by disrupting the non-covalent interactions and solubilizing the denatured proteins (Dehaghani et al., 2010).

To solubilize the above pellet, the cells were resuspended in a binding buffer containing 50 mM Tris base, 0.3 M NaCl, 8 M urea, and 20 mM imidazole (Condition 1 of solubilization). To achieve complete solubilization of the proteins, the suspension was incubated at room temperature or four °C for a certain period, typically 30 minutes to 1 hour. The optimal solubilization time was 16 hours, which depends on the specific protein and experimental

conditions. After solubilization, the insoluble debris was removed by centrifugation, and the resulting supernatant was collected for affinity chromatography. The protein concentration in the supernatant was determined using UV absorbance spectroscopy. Alternatively, the solubilization buffer was modified by adding 1% Triton X 100 and 1% Tween 20 along with 50 mM Tris base, 0.3 M NaCl, 8 M urea, and 20 mM imidazole (Condition 2 of solubilization) resulting in a binding buffer variation termed Binding buffer 2. This modified buffer showed increased success in solubilizing the target proteins, providing an enhanced solubilization efficiency for downstream purification processes.

3.2.7 Ni-NTA Chromatography

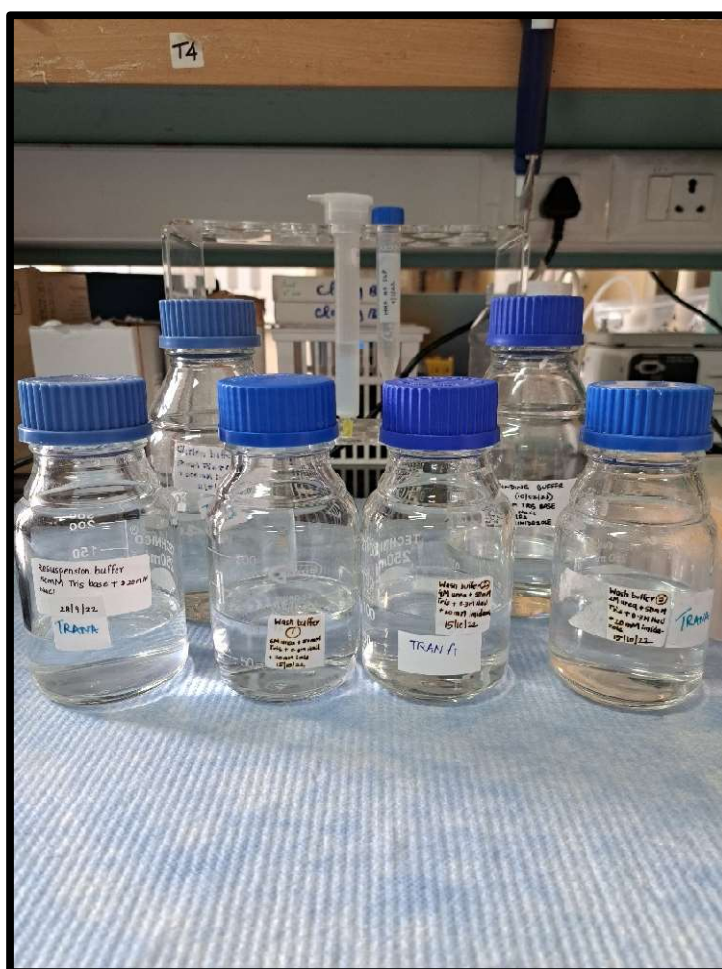


Figure 3.2.7.1 Purification Set-up with Buffers and Test Sample

Ni-NTA chromatography utilizes a nickel-charged resin to selectively bind the His-tagged target protein. By exploiting the affinity of the His-tag for nickel, the desired protein can be purified from the crude lysate, while other contaminants are washed away (Harrison, 1993).

To purify the solubilized protein sample obtained from the above step, Ni-NTA chromatography was performed. The Ni-NTA resin was equilibrated with a binding buffer containing 50 mM Tris base, 0.3 M NaCl, 8 M urea, and 20 mM imidazole. The column was loaded with the solubilized protein sample, and the unbound proteins were washed off using a wash buffer containing 50 mM Tris base, 0.3 M NaCl, 20 mM imidazole, and decreasing concentrations of urea (6 M, 4 M, and 2 M). The target protein was then eluted from the column using an elution buffer containing 50 mM Tris base, 0.3 M NaCl, and 500 mM imidazole. The eluted fractions obtained were subjected to analysis by SDS-PAGE, and the protein concentration was determined using UV absorbance spectroscopy.

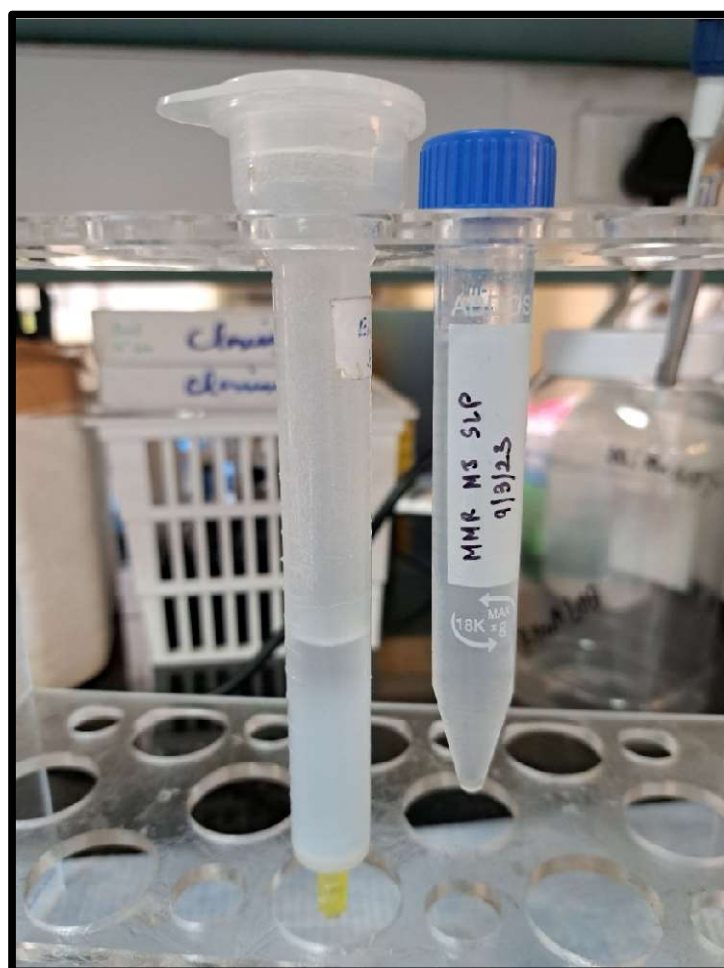


Figure 3.2.7.2 Ni-NTA Chromatography of M4

3.2.8 Dialysis

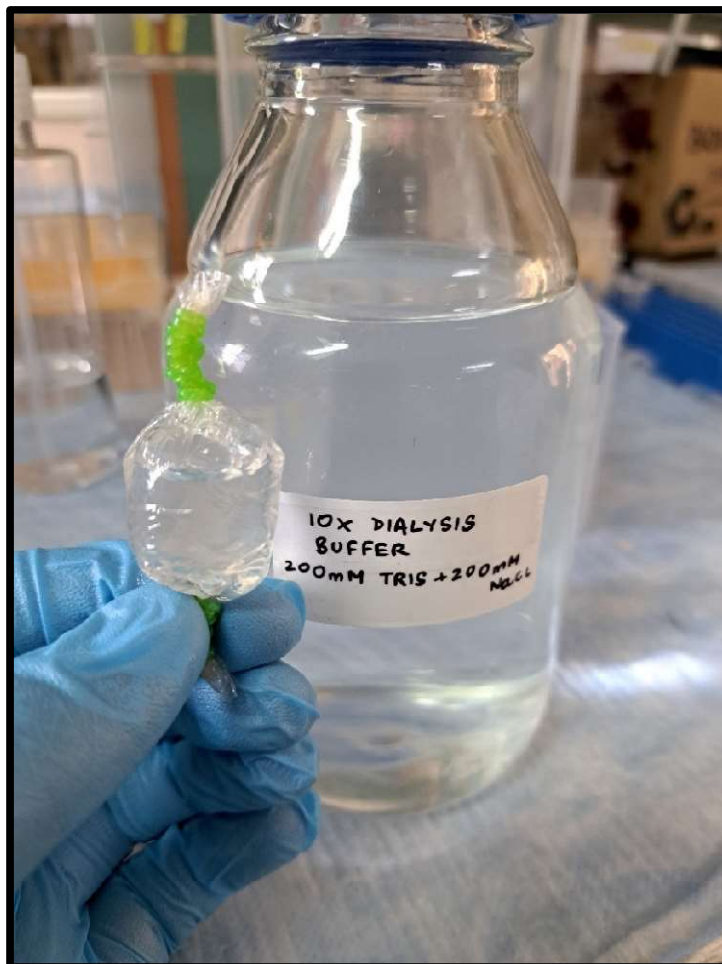


Figure 3.2.8.1 Dialysis Bag with Buffer

Dialysis involves the removal of small molecules and buffer components from the purified protein sample. The protein-containing solution is placed in a dialysis bag and immersed in a buffer, allowing for the exchange of unwanted substances and adjustment of the sample to the desired buffer conditions (Vikis, 2004).

After the purification of the target protein by Ni-NTA chromatography, the eluted fractions containing the protein were dialyzed against a buffer containing 20 mM (0.02 M) Tris base and 20 mM (0.02 M) NaCl using snake skin dialysis bags. The dialysis process was performed to remove any remaining imidazole or other small molecules and to exchange the buffer to the desired conditions for downstream applications. The Dialysis was carried out overnight. The effectiveness of the Dialysis was confirmed by analyzing the protein sample by SDS-PAGE and checking for the absence of any residual contaminants or buffer components. After Dialysis, the protein concentration was determined using UV absorbance spectroscopy, and the

protein sample was ready for downstream applications such as functional or structural characterization.

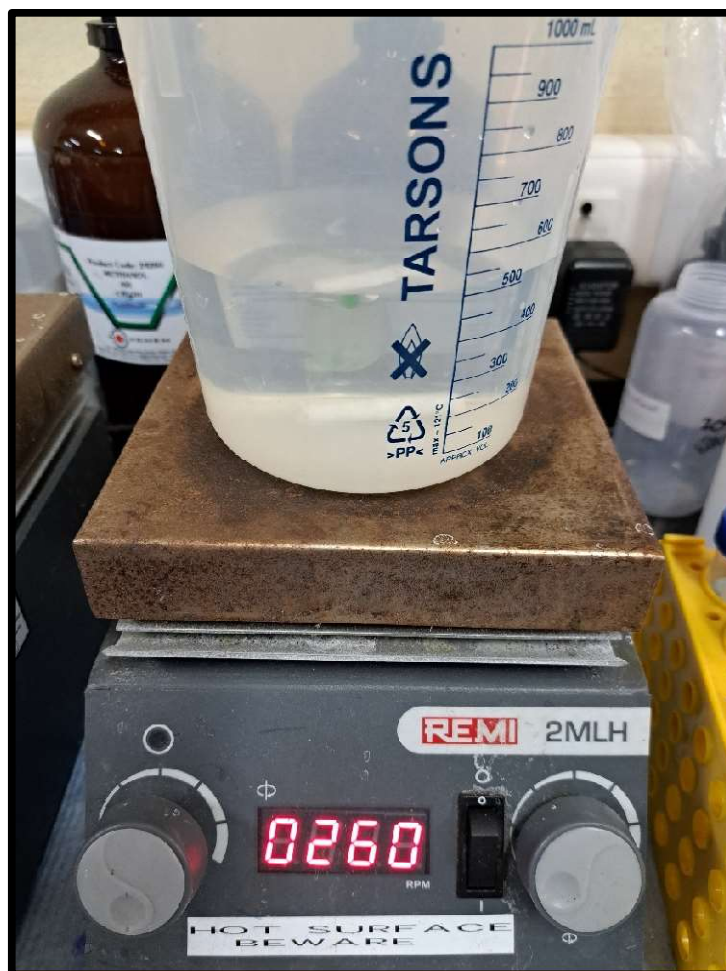


Figure 3.2.8.2 Dialysis of M4

3.2.9 Sugar Binding Affinity Chromatography

Sugar binding chromatography is used to assess the successful refolding of solubilized proteins after dialysis (Lebendiker and Danieli, 2017). The technique involves loading the protein sample onto a sugar column and selectively eluting the target protein using different elution buffers. Elution Buffer 1 (Condition 1 of sugar binding) contains Tris-Cl, NaCl, and EDTA, while Elution Buffer 2 (Condition 2 of sugar binding) includes additional components such as urea and β -mercaptoethanol. Elution Buffer 3 (Condition 3 of sugar binding) utilizes a sugar as an elution agent. By analyzing the eluted fractions, researchers can determine the protein's refolding efficiency and confirm the proper folding of its sugar binding domain.

3.3 Component specifications

- Resuspension buffer: 50mM Tris base + 20mM NaCl
 - Tris base is commonly used as a buffering agent in biological systems due to its ability to maintain a stable pH. The pH of the resuspension buffer is typically adjusted to a pH value that favours the stability of the protein of interest.
 - NaCl is added to the buffer to provide a physiological ionic strength, which is important for maintaining the solubility and stability of many proteins.
- Binding buffer 1 (Condition 1 of solubilization): 50mM Tris base + 0.3M NaCl + 8M urea + 20mM imidazole
 - Tris base and NaCl serve the same function as in the resuspension buffer.
 - Urea is a chaotropic agent that disrupts non-covalent interactions, such as hydrogen bonds and hydrophobic interactions, which can aid in the solubilization of proteins that are difficult to purify.
 - Imidazole is used as a ligand for immobilized metal ion affinity chromatography (IMAC) columns, which can selectively bind to Poly histidine (His) tags on recombinant proteins. This can be useful for purifying recombinant proteins that have been engineered to contain His tags.
- Binding buffer 2 (Condition 2 of solubilization): 50mM Tris base + 0.3M NaCl + 8M urea + 20mM imidazole + 1% Tween 20 + 1% Triton X 100
 - Tween 20 (Polysorbate 20): Tween 20, a nonionic detergent, is used in binding buffer to reduce non-specific binding and minimize background noise in various assays.
 - Triton X-100 (Nonionic Surfactant): Triton X-100, a nonionic surfactant, is included in the binding buffer to disrupt hydrophobic interactions, prevent non-specific binding, and maintain solubility and stability of proteins and other biomolecules during the assay.
- Wash buffer 1-3 for Ni-NTA Chromatography: 50mM Tris base + 0.3M NaCl + 6-2M urea + 20mM imidazole
 - These buffers serve to remove non-specifically bound proteins and contaminants from the IMAC column.
 - The decreasing concentration of urea in each wash buffer serves to gradually reduce the strength of the interaction between the His-tagged protein and the IMAC column, which can help to minimize the loss of the target protein during the washing step.
- Elution buffer for Ni-NTA Chromatography: 50mM Tris base + 0.3M NaCl + 500mM imidazole

- The high concentration of imidazole in this buffer competes with the His-tagged protein for binding to the IMAC column, which results in the elution of the target protein.
 - The Tris base and NaCl serve the same function as in the previous buffers.
- Dialysis buffer: 20mM (0.02M) Tris base + 20mM (0.02M) NaCl
 - Dialysis is a commonly used method for removing low-molecular-weight contaminants from purified proteins.
 - The dialysis buffer typically contains a low salt concentration, which can help to prevent aggregation and precipitation of the protein during the dialysis process.
 - The Tris base serves as a buffering agent to maintain a stable pH during Dialysis.
- Elution Buffer 1 for Sugar Binding Chromatography (Condition 1 of sugar binding): 25mM Tris-Cl + 150mM NaCl + 2.5mM EDTA
 - Tris-Cl: Tris serves as a buffering agent to maintain a stable pH during elution.
 - NaCl: Sodium chloride is added to maintain ionic strength and support proper protein solubility during elution.
 - EDTA: Ethylenediaminetetraacetic acid (EDTA) is a chelating agent that helps remove metal ions, which can interfere with protein stability and purification.
- Elution Buffer 2 for Sugar Binding Chromatography (Condition 2 of sugar binding): 25mM Tris-Cl + 150mM NaCl + 2.5mM EDTA + 6M Urea + 10mM β -Mercaptoethanol
 - Urea: Urea is a chaotropic agent that disrupts non-covalent interactions, aiding in the solubilization of proteins and promoting their release from the resin.
 - β -Mercaptoethanol: β -Mercaptoethanol (β -ME) is a reducing agent that helps break disulfide bonds, which can contribute to protein aggregation and insolubility. By reducing disulfide bonds, β -ME enhances protein solubility and improves elution efficiency.
- Elution Buffer 3 for Sugar Binding Chromatography (Condition 3 sugar binding): 25mM Tris-Cl + 150mM NaCl + 2.5mM EDTA + 1M Sugar
 - A sugar molecule that is included as a competitive eluent to displace the M4 protein from the sugar affinity resin. Its high concentration helps effectively compete with the protein for binding sites, facilitating the release of the protein.
- Kanamycin: 50 μ g/ml- Kanamycin is added to M4 medium as an antibiotic to selectively inhibit the growth of bacteria that do not contain the kanamycin

resistance gene, allowing for the selection and maintenance of transformed or genetically modified bacterial cells.

- IPTG: 0.5mM- IPTG is an inducer that is added to the growth medium to activate the lac operon in bacterial cells. It binds to the lac repressor protein, preventing it from binding to the lac operator region, thereby inducing the expression of target genes under the control of the lac promoter.
- LB broth: 6.25g for 250 ml- LB broth is a nutrient-rich medium used for the cultivation of bacteria. It provides all the essential nutrients, such as amino acids, sugars, and salts, necessary for bacterial growth and proliferation. LB broth supports the growth of a wide range of bacterial species and is commonly used in microbiology research and molecular biology techniques, such as cloning and protein expression.

Table 3.3.1 List of equipment used during this project.

Equipment	Specifications
Eppendorf Centrifuge	Eppendorf centrifuge is a laboratory equipment used for the separation of components of a sample based on their density by spinning it at high speeds. It is used for pelleting bacterial cells, removing cell debris or isolation of protein aggregates after lysis.
BSC class 1	Biosafety Cabinet (BSC) is a type of containment equipment designed to protect the user and the environment from harmful biological agents. Class 1 BSC is used for handling low to moderate-risk biological agents, typically bacteria and viruses. It is used during bacterial cell culturing and manipulation, where it provides a sterile environment, prevents contamination, and protects the operator from exposure to pathogenic bacteria.
Branson Sonicator	Branson Sonicator is an ultrasonic homogenizer that uses high-frequency sound waves to disrupt cells and subcellular organelles. It is used for cell lysis and protein extraction from bacterial cells by disrupting cell membranes.
Ni-NTA chromatography column	Affinity chromatography using a Ni-NTA column is utilized for purifying recombinant proteins that have been genetically modified with an affinity tag, typically a His-tag, from bacterial cell lysates. A resin with a high affinity for the His-tag is packed into the column, enabling selective binding of the recombinant protein to the column.
Magnetic stirrer	The magnetic stirrer is laboratory equipment used for stirring liquid solutions. It is used during protein purification

	steps such as buffer exchange, Dialysis, or column chromatography to mix solutions and ensure that the buffer components are uniformly mixed.
BioBee Tube rotator	BioBee tube rotator is laboratory equipment used for rotating tubes containing biological samples. It is used for the gentle mixing of bacterial cell cultures and protein solutions, avoiding shear forces that can damage the protein structure.
Thermofisher Nanodrop	Thermofisher Nanodrop is a spectrophotometer used for measuring the concentration and purity of nucleic acids and proteins. It is used for determining the concentration of bacterial cell lysates and purified recombinant protein solutions and to check for contamination with nucleic acids or other proteins.

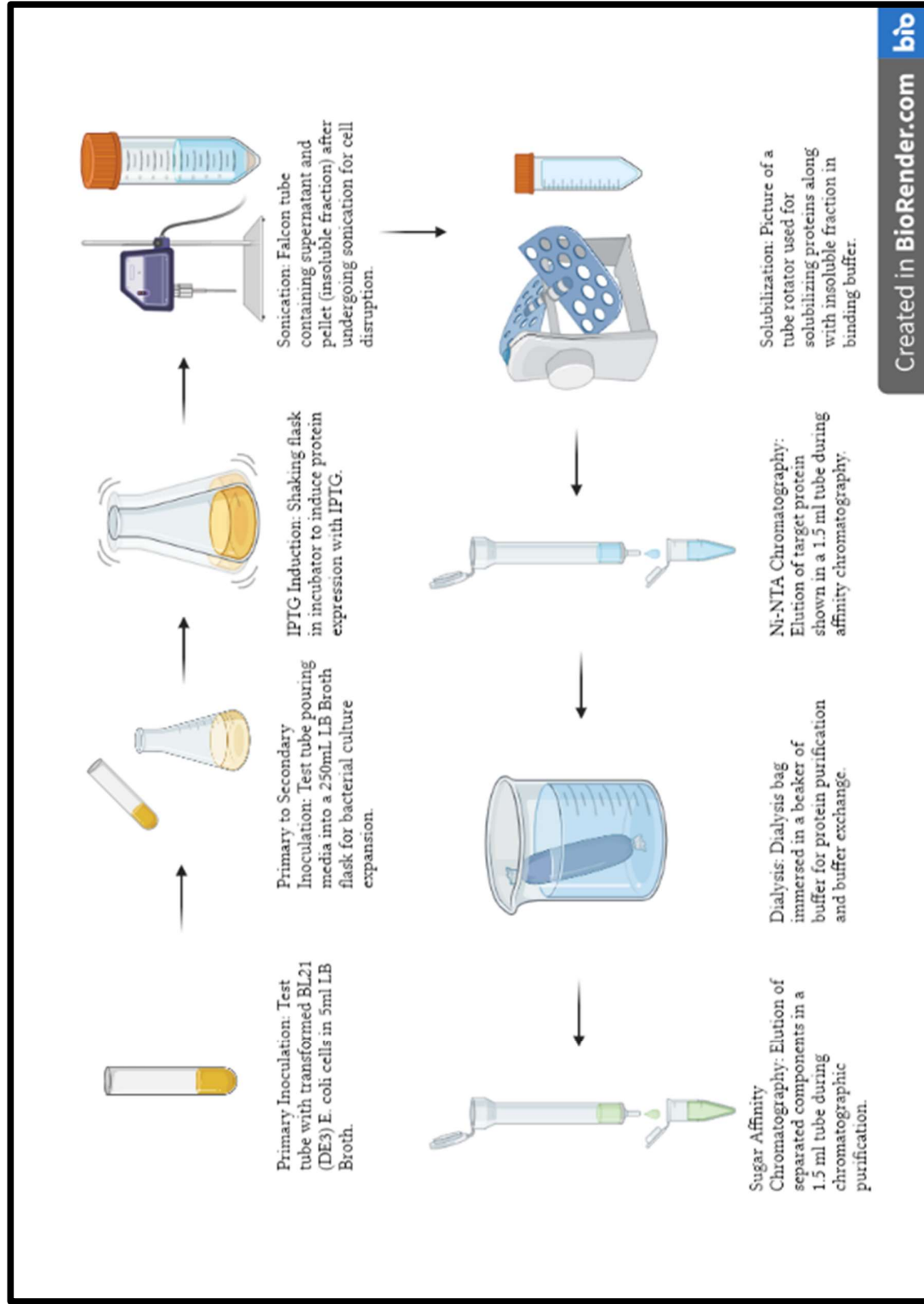


Figure 3.3.1 An illustration of the steps involved in expression and purification of M4.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results and analysis of the research study conducted. The primary objective of this chapter is to provide a comprehensive overview of the findings obtained from the experimental data and their significance in relation to the research objectives. The chapter begins with an introduction that outlines the key aspects to be discussed, followed by a detailed analysis of the results presented in graphical or tabular form. The interpretation and explanation of the results will be provided, highlighting their implications and relevance to the research questions. Furthermore, any deviations from the expected results will be addressed, and justifications for these deviations will be provided based on the experimental setup or other factors.

4.2 Result analysis

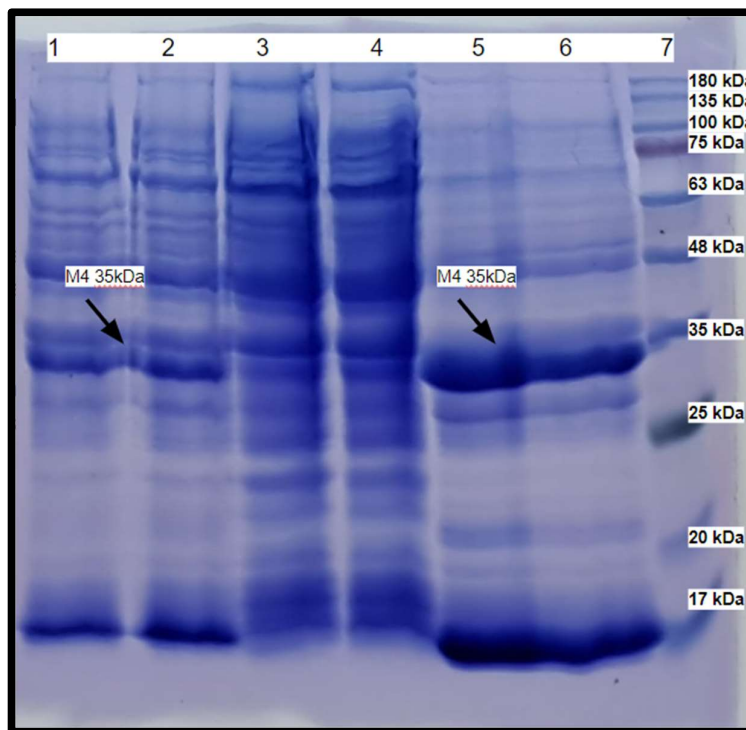


Figure 4.2.1 Expression of M4 using IPTG BL21 (DE3)

Loading Pattern:

1. Post IPTG induction
2. Post IPTG induction

3. Sonication Supernatant
4. Sonication Supernatant
5. Sonication Pellet
6. Sonication Pellet
7. Pre-stained Protein Ladder (270-6.5kDa)

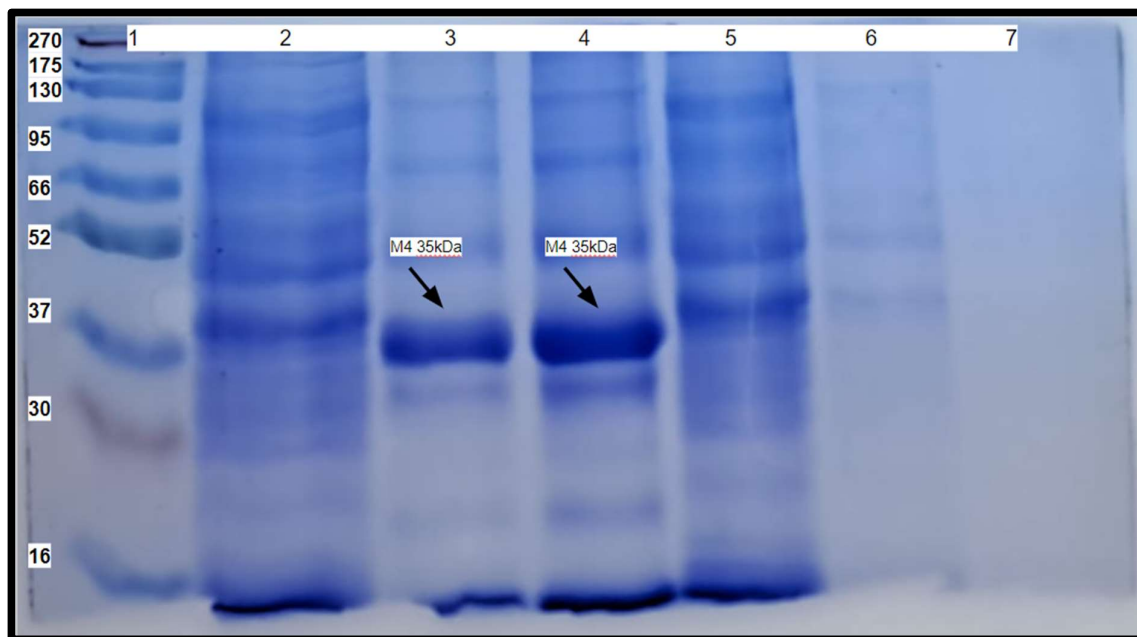


Figure 4.2.2 Solubilization of M4 under Conditions 1 and 2 of Solubilization (refer 3.2.6)

Loading Pattern:

1. Pre-stained Protein Ladder (270-6.5kDa)
2. Solubilized Supernatant under Condition 1.
3. Solubilized Pellet under Condition 1
4. Solubilized Supernatant under Condition 2
5. Solubilized Pellet under Condition 2

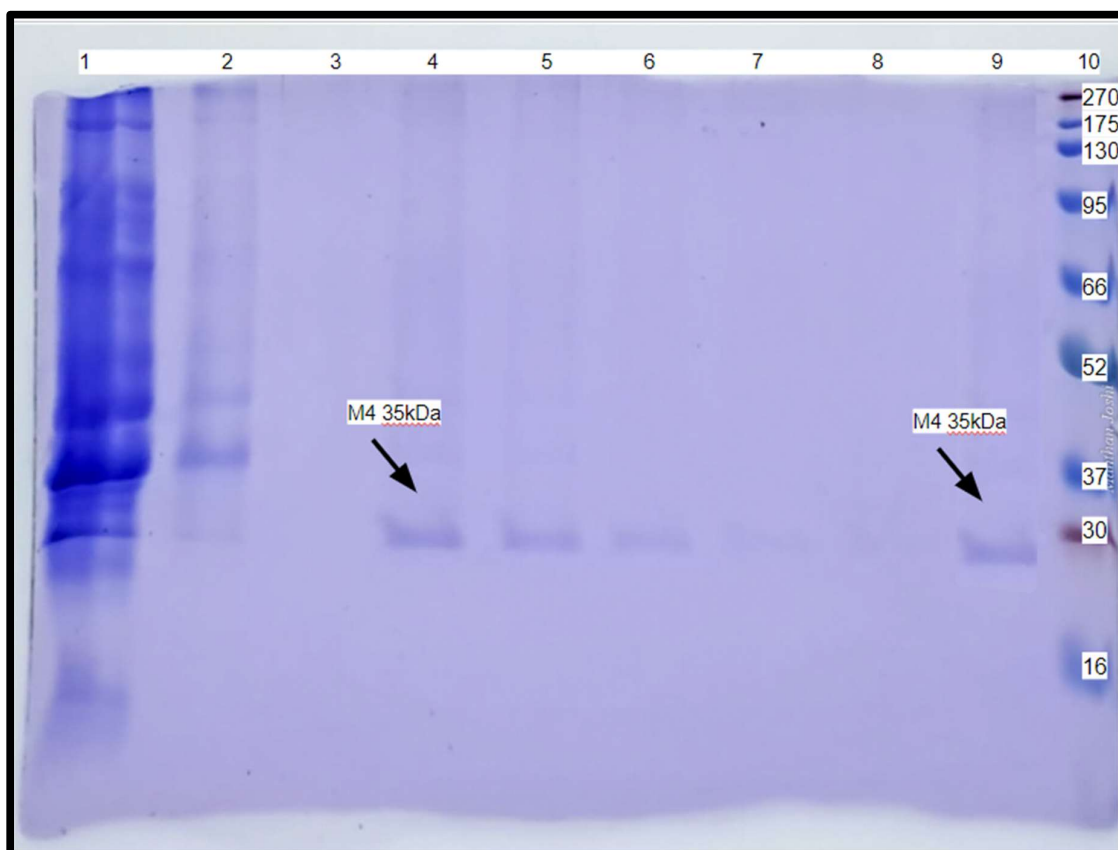


Figure 4.2.3 Ni-NTA Chromatography Purification of M4 under Conditions 1 and 2 of Solubilization (refer 3.2.6)

Loading Pattern:

1. Flowthrough
2. Wash 1
3. Wash 3
4. Elute under Condition 2.
5. Elute under Condition 2.
6. Elute under Condition 1.
7. Elute under Condition 1.
8. Dialysis under Condition 1.
9. Dialysis under Condition 2.
10. Pre-stained Protein Ladder (250-6.5 kDa)

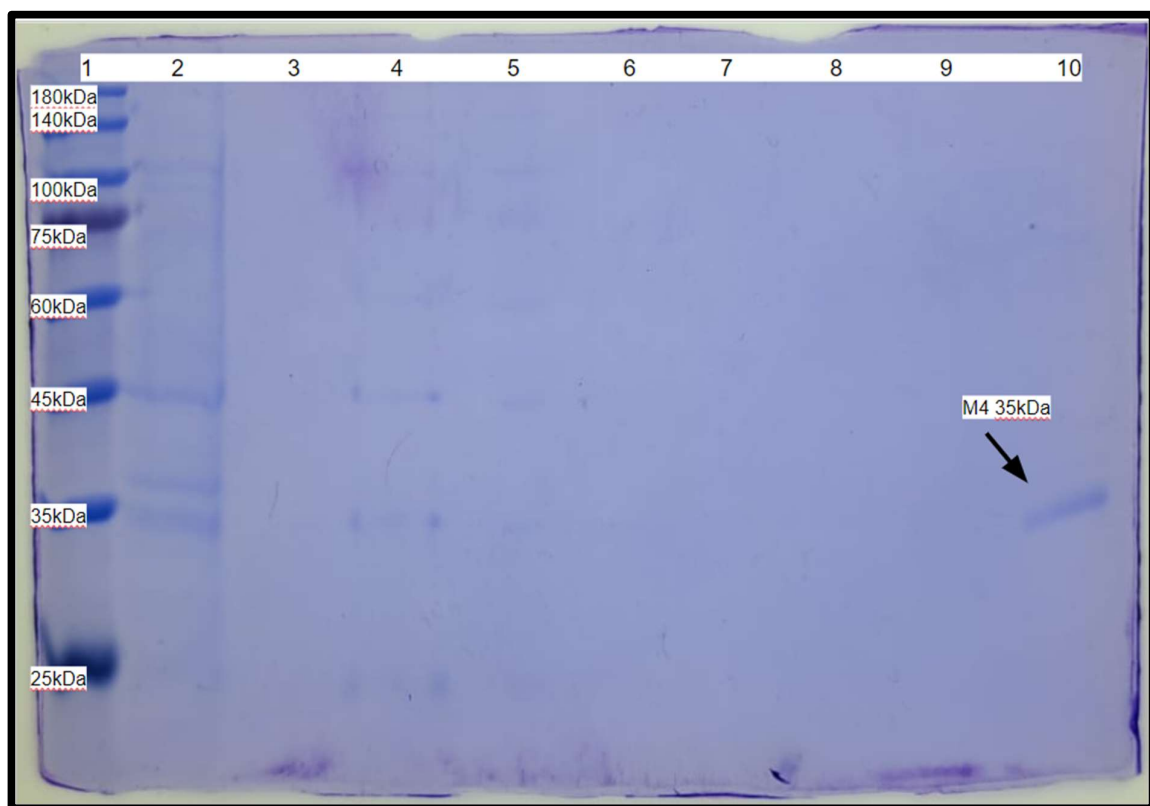


Figure 4.2.4 Sugar Binding Chromatography for M4 under Conditions 1, 2 and 3 (refer 3.2.9)

Loading Pattern

1. Pre-stained Protein Ladder (260-6.5 kDa)
2. Flowthrough
3. Wash 3
4. Wash 5
5. Elute (Standardized elution buffer)
6. Elute (Standardized elution buffer)
7. Elute (Elution buffer 6M Urea and 10mM Beta Amine)
8. Elute (Elution buffer 6M Urea and 10mM Beta Amine)
9. Elute (Elution buffer 1M Sugar)
10. Elute (Elution buffer 1M Sugar)

4.3 Significance of the Results Obtained

In order to assess the concentration of M4, the solubilized protein was analyzed using spectrophotometry at 280 nm and converted into protein concentration. The dialyzed M4, solubilized in the binding buffer of 8M urea, exhibited an absorbance at 280 nm of 0.46, while the binding buffer containing 1% triton X 100 and 1% tween 20 showed an absorbance of 0.68.

By applying the formula $\text{Protein concentration (mg/ml)} = (\text{Absorbance at 280 nm}) / (\text{Extinction coefficient at 280 nm (95965)} \times \text{Path length in cm (10mm)})$, the protein concentrations were calculated. For the 8M urea buffer, the concentration of M4 was determined to be approximately 0.000048 mg/ml, which corresponds to around 1.37 μM (47.95 ng/ μl) when considering the molecular weight of M4 (35 kDa). Similarly, for the buffer containing triton X 100 and tween 20, the concentration was found to be approximately 0.000071 mg/ml, equivalent to around 2.03 μM (71.05 ng/ μl). These results provide valuable insights into the concentration of M4 in the respective buffers, offering crucial information for the interpretation and design of subsequent experiments and analyses.

The results obtained from the solubilization experiments using different binding buffers have significant implications for the purification of the M4 protein (35kDa mol. wt). The observation that solubilization with Binding Buffer 2 (50mM Tris base + 0.3M NaCl + 8M urea + 20mM imidazole + 1% Triton X 100 + 1% Tween 20) resulted in higher A280 values compared to Binding Buffer 1 (50mM Tris base + 0.3M NaCl + 8M urea + 20mM imidazole) indicates that the inclusion of Triton X 100 and Tween 20 enhances the solubilization efficiency. This finding suggests that the addition of these detergents, along with 8M urea, promotes the effective solubilization of the M4 protein (35kDa mol. wt.), potentially improving the downstream purification processes.

Moreover, the higher protein yield observed in the Ni-NTA chromatography and dialysis steps when solubilized with Binding Buffer 2 further emphasizes the significance of this modified buffer. The increased elution of the target protein suggests that the inclusion of Triton X 100 and Tween 20 in the solubilization buffer improves the protein's affinity for the Ni-NTA resin, facilitating its purification. The components of Binding Buffer 2, including 50mM Tris base, 0.3M NaCl, 8M urea, 20mM imidazole, 1% Triton X 100, and 1% Tween 20, collectively contribute to enhancing solubilization and purification efficiency.

In the sugar affinity chromatography experiment, the use of different elution buffers (Elution Buffer, Elution Buffer 2, and Elution Buffer 3) did not result in successful elution of the M4 protein, indicating a strong binding affinity between the protein and the resin. Notably, a single elution with 1M sugar revealed a faint 35kDa protein band in the SDS PAGE, indicating that the sugar was able to bind to the resin. However, the tightness of this binding suggests that alternative elution methods should be explored to effectively release the M4 protein from the resin.

4.4 Deviation from Expected Results and Justification

The observation that the M4 protein did not effectively elute in the sugar affinity chromatography experiments, despite the inclusion of 1M sugar in the elution buffer, deviates from the expected results. Notably, a faint 35kDa protein band was observed in the SDS PAGE, indicating that the sugar has bound to the resin, albeit with a seemingly tight interaction. The high sugar content in the elution buffer poses challenges for subsequent purification steps, necessitating the need for effective dialysis to remove the excess sugar.

One possible justification for this deviation is the presence of alternative or stronger binding interactions between the M4 protein and the resin that are not easily disrupted by the tested elution conditions. These interactions may involve additional binding sites or non-specific interactions, resulting in an overall stronger binding affinity. Despite the inclusion of components such as 25mM Tris-Cl, 150mM NaCl, 2.5mM EDTA, 6M Urea, and 10mM b-Mercaptoethanol in the elution buffer, the protein-resin interaction remains complex and resistant to disruption.

These findings highlight the unique properties of the M4 protein and the need for alternative strategies to overcome the strong resin binding. Moreover, the high sugar content necessitates effective dialysis to remove it, which can be a challenging task. This observation provides valuable insights for further studies aiming to improve the solubilization, purification, and characterization of the M4 protein, emphasizing the importance of detergent additives for efficient solubilization and underscoring the need for innovative approaches in the presence of strong resin binding.

CHAPTER 5

CONCLUSIONS AND FUTURE SCOPE OF WORK

5.1 Brief Summary of the Work

The objective of this study was to investigate the solubilization and purification of the M4 protein and evaluate the effectiveness of different buffers and chromatographic techniques. The methodology involved the use of binding buffers with varying compositions, including the addition of detergents Triton X 100 and Tween 20, as well as sugar affinity chromatography. The experimental results demonstrated that solubilization with Binding Buffer 2, containing Triton X 100 and Tween 20, resulted in higher protein yield during the purification steps. However, the sugar affinity chromatography did not yield successful elution of the M4 protein, indicating limitations in the traditional approach.

5.2 Conclusions

Based on the findings of this study, several conclusions can be drawn. Firstly, the inclusion of Triton X 100 and Tween 20 in the solubilization buffer (Binding Buffer 2) significantly enhanced the solubilization efficiency and overall protein yield. This observation aligns with previous studies that have reported the role of detergents in improving the solubility and stability of recombinant proteins (Eshaghi et al., 2005; Chipot et al., 2018). The higher A280 values obtained with Binding Buffer 2 also indicate better protein solubility and quality, which is crucial for downstream purification processes.

The unsuccessful elution of the M4 protein in the sugar affinity chromatography experiments, despite the inclusion of various elution buffers, underscores the limitations of this technique. Notably, a faint 35kDa protein band observed in the SDS PAGE confirmed the binding of the sugar to the resin, albeit with a remarkably strong interaction. Removing the high concentration of 1M sugar from the elution would pose challenges, necessitating effective dialysis to remove the excess sugar. These findings align with previous studies reporting difficulties in eluting proteins with strong binding affinities to affinity resins (Burgess-Brown et al., 2008; DeLaBarre et al., 2011). Therefore, alternative approaches should be explored to address the unique binding characteristics of the M4 protein. Future investigations should consider innovative elution strategies that can overcome the tight resin binding and ensure successful purification of the M4 protein. Additionally, the implementation of efficient dialysis techniques will be crucial to remove the excessive sugar content and advance downstream characterization of the purified protein.

5.3 Future Scope of Work

Based on the limitations encountered in this study, there are several future directions that can be pursued to further enhance the solubilization and purification of the M4 protein. These include:

- 5.3.1 *Exploring alternative elution techniques:* Given the strong binding affinity of the M4 protein to the sugar affinity resin, it would be beneficial to investigate alternative elution methods that can effectively release the target protein from the resin while maintaining its stability and functionality. For example, different elution buffers or conditions could be explored, such as varying the pH, salt concentration, or introducing competitive ligands specific to the M4 protein. These alternative elution techniques may provide improved purification outcomes and enhance the overall solubilization and purification strategy (Ayyar et al, 2012; Urh et al., 2009).
- 5.3.2 *Protein engineering and tag optimization:* Protein engineering approaches, such as site-directed mutagenesis, can be employed to introduce specific amino acid substitutions or modifications in the M4 protein. This may alter the binding affinity to the affinity resin and facilitate efficient elution. Additionally, optimizing the design and position of affinity tags, such as the His-tag, can enhance the specificity and selectivity of purification techniques (Ott et al., 2017). Rational tag optimization can contribute to improved solubility, binding, and purification efficiency.
- 5.3.3 *Investigation of alternative affinity ligands:* Instead of sugar, exploring other affinity ligands that can selectively bind to the M4 protein may offer new opportunities for its purification. This could involve screening a range of sugars or small molecules that exhibit affinity towards the protein of interest. Identifying a ligand with higher binding affinity and specific recognition properties would enable the development of a more efficient affinity chromatography method for purification (Reese et al., 2020).

REFERENCES

1. F. Baneyx, "Recombinant protein expression in *Escherichia coli*," Current opinion in biotechnology, vol. 10, no. 5, pp. 411-421, 1999.
2. D. M. Francis and R. Page, "Strategies to optimize protein expression in *E. coli*," Current protocols in protein science, vol. 61, no. 1, pp. 5-24, 2010.
3. K. B. Barken, J. A. Haagensen, and T. Tolker-Nielsen, "Advances in nucleic acid-based diagnostics of bacterial infections," Clinica Chimica Acta, vol. 384, no. 1-2, pp. 1-11, 2007.
4. J. H. Choi and S. Lee, "Secretory and extracellular production of recombinant proteins using *Escherichia coli*," Applied microbiology and biotechnology, vol. 64, pp. 625-635, 2004.
5. J. Sambrook, D. W. Russell, and J. Sambrook, Molecular cloning: A laboratory manual, vol. 3, Cold Spring Harbor Laboratory Press, 2001.
6. F. M. Wurm, "Production of recombinant protein therapeutics in cultivated mammalian cells," Nature biotechnology, vol. 22, no. 11, pp. 1393-1398, 2004.
7. V. Raja and K. Natarajaseenivasan, "Pathogenic, diagnostic and vaccine potential of leptospiral outer membrane proteins (OMPs)," Critical reviews in microbiology, vol. 41, no. 1, pp. 1-17, 2015.
8. F. Cedrone, A. Ménez, and E. Quéméneur, "Tailoring new enzyme functions by rational redesign," Current opinion in structural biology, vol. 10, no. 4, pp. 405-410, 2000.
9. B. R. Glick and C. L. Patten, Molecular biotechnology: principles and applications of recombinant DNA. John Wiley & Sons, 2022.
10. G. Walsh, "Biopharmaceutical benchmarks 2018," Nature Biotechnology, vol. 36, no. 12, pp. 1136-1145, 2018.
11. E. G. Cormier, D. N. Tran, L. Yukhayeva, W. C. Olson, and T. Dragic, "Mapping the determinants of the CCR5 amino-terminal sulfopeptide interaction with soluble human immunodeficiency virus type 1 gp120-CD4 complexes," Journal of virology, vol. 75, no. 12, pp. 5541-5549, 2001.
12. N. C. Shaner, R. E. Campbell, P. A. Steinbach, B. N. Giepmans, A. E. Palmer, and R. Y. Tsien, "Improved monomeric red, orange and yellow fluorescent proteins derived from *Discosoma* sp. red fluorescent protein," Nature Biotechnology, vol. 22, no. 12, pp. 1567-1572, 2005.
13. C. Ferndahl, N. Bonander, C. Logez, R. Wagner, L. Gustafsson, C. Larsson et al., "Increasing cell biomass in *Saccharomyces cerevisiae* increases recombinant protein yield: the use of a respiratory strain as a microbial cell factory," Microbial cell factories, vol. 9, no. 1, pp. 1-11, 2010.

14. T. Unger and Y. Peleg, "Recombinant protein expression in the baculovirus-infected insect cell system," *Chemical Genomics and Proteomics: Reviews and Protocols*, pp. 187-199, 2012.
15. E. Mills, I. S. Pultz, H. D. Kulasekara, and S. I. Miller, "The bacterial second messenger c-di-GMP: mechanisms of signalling," *Cellular microbiology*, vol. 13, no. 8, pp. 1122-1129, 2011.
16. S. Macauley-Patrick, M. L. Fazenda, B. McNeil, and L. M. Harvey, "Heterologous protein production using the *Pichia pastoris* expression system," *Yeast*, vol. 22, no. 4, pp. 249-270, 2005.
17. P. Stolt-Bergner, C. Benda, T. Bergbrede, H. Besir, P. H. Celie, C. Chang et al., "Baculovirus-driven protein expression in insect cells: A benchmarking study," *Journal of structural biology*, vol. 203, no. 2, pp. 71-80, 2018.
18. G. L. Rosano and E. A. Ceccarelli, "Recombinant protein expression in *Escherichia coli*: advances and challenges," *Frontiers in Microbiology*, vol. 5, p. 172, 2014.
19. H. P. Sørensen and K. K. Mortensen, "Advanced genetic strategies for recombinant protein expression in *Escherichia coli*," *Journal of Biotechnology*, vol. 115, no. 2, pp. 113-128, 2005.
20. K. Terpe, "Overview of tag protein fusions: from molecular and biochemical fundamentals to commercial systems," *Applied Microbiology and Biotechnology*, vol. 60, no. 5, pp. 523-533, 2003.
21. M. Berkmen, "Production of disulfide-bonded proteins in *Escherichia coli*," *Protein Expression and Purification*, vol. 82, no. 2, pp. 240-251, 2012.
22. J. Arnau, C. Lauritzen, G. E. Petersen, and J. Pedersen, "Current strategies for the use of affinity tags and tag removal for the purification of recombinant proteins," *Protein expression and purification*, vol. 48, no. 1, pp. 1-13, 2006.
23. K. Huse, H. J. Böhme, and G. H. Scholz, "Purification of antibodies by affinity chromatography," *Journal of biochemical and biophysical methods*, vol. 51, no. 3, pp. 217-231, 2002.
24. H. Liu, B. Zhang, and M. Yan, "Purification of recombinant proteins using molecular recognition technology," *Protein Expression and Purification*, vol. 95, pp. 104-109, 2014.
25. S. M. Singh and A. K. Panda, "Solubilization and refolding of bacterial inclusion body proteins," *Journal of bioscience and bioengineering*, vol. 99, no. 4, pp. 303-310, 2005.
26. D. L. Nelson and M. M. Cox, *Lehninger principles of biochemistry*, 4th ed. W. H. Freeman and Company, 2005.

27. H. Fabian and D. Naumann, Eds., Protein folding and misfolding: Shining light by infrared spectroscopy. Springer Science & Business Media, 2011.
28. S. Costa, A. Almeida, A. Castro, and L. Domingues, "Fusion tags for protein solubility, purification and immunogenicity in *Escherichia coli*: the novel Fh8 system," *Frontiers in microbiology*, vol. 5, p. 63, 2014.
29. J. Chico, A. Rúbies, F. Centrich, R. Companyó, M. D. Prat, and M. Granados, "Use of gel permeation chromatography for clean-up in the analysis of coccidiostats in eggs by liquid chromatography–tandem mass spectrometry," *Analytical and bioanalytical chemistry*, vol. 405, pp. 4777-4786, 2013.
30. D. S. Hage, "High-performance affinity chromatography: a powerful tool for studying serum protein binding," *Journal of Chromatography B*, vol. 768, no. 1, pp. 3-30, 2002.
31. Villaverde and M. Mar Carrió, "Protein aggregation in recombinant bacteria: biological role of inclusion bodies," *Biotechnology letters*, vol. 25, pp. 1385-1395, 2003.
32. Shukla and J. Thömmes, "Recent advances in large-scale production of monoclonal antibodies and related proteins," *Trends in Biotechnology*, vol. 28, no. 5, pp. 253-261, 2010.
33. H. P. Sørensen and K. K. Mortensen, "Soluble expression of recombinant proteins in the cytoplasm of *Escherichia coli*," *Microbial Cell Factories*, vol. 4, pp. 1-8, 2005.
34. M. Carrió, N. González-Montalbán, A. Vera, A. Villaverde, and S. Ventura, "Amyloid-like properties of bacterial inclusion bodies," *Journal of molecular biology*, vol. 347, no. 5, pp. 1025-1037, 2005.
35. M. M. Carrió and A. Villaverde, "Protein aggregation as bacterial inclusion bodies is reversible," *FEBS letters*, vol. 489, no. 1, pp. 29-33, 2001.
36. Taglialegna, I. Lasa, and J. Valle, "Amyloid structures as biofilm matrix scaffolds," *Journal of bacteriology*, vol. 198, no. 19, pp. 2579-2588, 2016.
37. P. P. De Laureto et al., "Protein aggregation and amyloid fibril formation by an SH3 domain probed by limited proteolysis," *Journal of molecular biology*, vol. 334, no. 1, pp. 129-141, 2003.
38. Y. L. Luo et al., "Macrophage-specific in vivo gene editing using cationic lipid-assisted polymeric nanoparticles," *ACS nano*, vol. 12, no. 2, pp. 994-1005, 2018.
39. G. Zago, P. H. Saavedra, K. R. Keshari, and J. S. Perry, "Immunometabolism of tissue-resident macrophages—an appraisal of the current knowledge and cutting-edge methods and technologies," *Frontiers in Immunology*, vol. 12, p. 665782, 2021.
40. S. Bakeera-Kitaka, N. Nabukeera-Barungi, C. Nöstlinger, K. Addy, and R. Colebunders, "Sexual risk reduction needs of adolescents living with HIV in a clinical care setting," *AIDS care*, vol. 20, no. 4, pp. 426-433, 2008.

41. Y. Pei and Y. Yeo, "Drug delivery to macrophages: challenges and opportunities," *Journal of Controlled Release*, vol. 240, pp. 202-211, 2016.
42. E. Hochuli, H. Dobeli, and A. Schacher, "New metal chelate adsorbent selective for proteins and peptides containing neighboring histidine residues," *Journal of Chromatography A*, vol. 411, pp. 177-184, 1987.
43. R. B. Kapust, J. Tözsér, T. D. Copeland, and D. S. Waugh, "The P1' specificity of tobacco etch virus protease," *Biochemical and Biophysical Research Communications*, vol. 294, no. 5, pp. 949-955, 2002.
44. R. C. F. Cheung, J. H. Wong, and T. B. Ng, "Immobilized metal ion affinity chromatography: a review on its applications," *Applied microbiology and biotechnology*, vol. 96, pp. 1411-1420, 2012.
45. T. G. Schmidt and A. Skerra, "One-step affinity purification of bacterially produced proteins by means of the 'Strep tag' and immobilized recombinant core streptavidin," *Journal of Chromatography A*, vol. 676, no. 1-2, pp. 337-345, 1994.
46. D. Esposito and D. K. Chatterjee, "Enhancement of soluble protein expression through the use of fusion tags," *Current Opinion in Biotechnology*, vol. 17, no. 4, pp. 353-358, 2006.
47. J. Carpousis, "The RNA degradosome of *Escherichia coli*: an mRNA-degrading machine assembled on RNase E," *Annu. Rev. Microbiol.*, vol. 61, pp. 71-87, 2007.
48. T. K. Singha et al., "Efficient genetic approaches for improvement of plasmid based expression of recombinant protein in *Escherichia coli*: A review," *Process Biochemistry*, vol. 55, pp. 17-31, 2017.
49. S. Hussain, "Dose-response relationships for mutations induced in *E. coli* by some model compounds," *Hereditas*, vol. 101, no. 1, pp. 57-65, 1984.
50. S. Okuda and H. Tokuda, "Model of mouth-to-mouth transfer of bacterial lipoproteins through inner membrane LolC, periplasmic LolA, and outer membrane LolB," *Proceedings of the National Academy of Sciences*, vol. 106, no. 14, pp. 5877-5882, 2009.
51. J. Ma et al., "Overexpression and characterization of a lipase from *Bacillus subtilis*," *Protein expression and purification*, vol. 45, no. 1, pp. 22-29, 2006.
52. H. Chen et al., "Strategies and applications of antigen-binding fragment (Fab) production in *Escherichia coli*," *Pharmaceutical Fronts*, vol. 3, no. 02, pp. e39-e49, 2021.
53. T. J. Park et al., "Cloning, expression, isotope labeling, purification, and characterization of bovine antimicrobial peptide, lactophorin in *Escherichia coli*," *Protein Expression and Purification*, vol. 65, no. 1, pp. 23-29, 2009.

54. Malygin et al., "Expression and purification of human ribosomal proteins S3, S5, S10, S19, and S26," *Protein expression and purification*, vol. 28, no. 1, pp. 57-62, 2003.
55. G. Candiano et al., "Blue silver: A very sensitive colloidal Coomassie G-250 staining for proteome analysis," *Electrophoresis*, vol. 25, no. 9, pp. 1327-1333, 2004.
56. U. K. Laemmli, "Cleavage of structural proteins during the assembly of the head of bacteriophage T4," *Nature*, vol. 227, no. 5259, pp. 680-685, 1970.
57. V. Neuhoff et al., "Improved staining of proteins in polyacrylamide gels including isoelectric focusing gels with clear background at nanogram sensitivity using Coomassie Brilliant Blue G-250 and R-250," *Electrophoresis*, vol. 9, no. 6, pp. 255-262, 1988.
58. Shevchenko et al., "Mass spectrometric sequencing of proteins silver-stained polyacrylamide gels," *Analytical Chemistry*, vol. 68, no. 5, pp. 850-858, 1996.
59. E. C. Nice, "The separation sciences, the front end to proteomics: An historical perspective," *Biomedical Chromatography*, vol. 35, no. 1, pp. e4995, 2021.
60. J. M. Walker, "The basics of Western blotting," in *The Protein Protocols Handbook*, Humana Press, 2005, pp. 445-451.
61. M. Vardakou et al., "Comparative analysis and validation of the malachite green assay for the high throughput biochemical characterization of terpene synthases," *MethodsX*, vol. 1, pp. 187-196, 2014.
62. J. S. Rosenblum et al., "Nuclear import and the evolution of a multifunctional RNA-binding protein," *The Journal of cell biology*, vol. 143, no. 4, pp. 887-899, 1998.
63. J. C. Cruz et al., "Conformational changes and catalytic competency of hydrolases adsorbing on fumed silica nanoparticles: II. Secondary structure," *Colloids and Surfaces B: Biointerfaces*, vol. 81, no. 1, pp. 1-10, 2010.
64. Schuler, "Single-molecule fluorescence spectroscopy of protein folding," *ChemPhysChem*, vol. 6, no. 7, pp. 1206-1220, 2005.
65. J. Carpousis, "The RNA degradosome of *Escherichia coli*: An mRNA-degrading machine assembled on RNase E," *Annual Review of Microbiology*, vol. 61, pp. 71-87, 2007.
66. Barakat Abdul Razzaq Muttar, Rakesh Kumar R. Panchal. "Overexpression of recombinant Staphylokinase and reduction of inclusion bodies using IPTG as an inducer in *E. coli* BL21 DE3." 03 August 2022. PREPRINT (Version 1) available at Research Square, doi: 10.21203/rs.3.rs-1912568/v1.
67. F. W. Studier et al., "Use of T7 RNA polymerase to direct expression of cloned genes," *Methods in Enzymology*, vol. 185, pp. 60-89, 1990.

68. L. Cao et al., "Production of soluble and active recombinant human thrombopoietin in *Escherichia coli* by fusing with SUMO," *Protein Expression and Purification*, vol. 143, pp. 53-59, 2018.
69. S. C. Gill and P. H. von Hippel, "Calculation of protein extinction coefficients from amino acid sequence data," *Analytical Biochemistry*, vol. 182, no. 2, pp. 319-326, 1989.
70. N. Pace et al., "How to measure and predict the molar absorption coefficient of a protein," *Protein Science*, vol. 4, no. 11, pp. 2411-2423, 1995.
71. J. C. Joly et al., "Overexpression of *Escherichia coli* oxidoreductases increases recombinant insulin-like growth factor-I accumulation," *Proceedings of the National Academy of Sciences*, vol. 95, no. 6, pp. 2773-2777, 1998.
72. Hanahan, "Studies on transformation of *Escherichia coli* with plasmids," *Journal of Molecular Biology*, vol. 166, no. 4, pp. 557-580, 1983.
73. W. Studier and B. A. Moffatt, "Use of bacteriophage T7 RNA polymerase to direct selective high-level expression of cloned genes," *Journal of Molecular Biology*, vol. 189, no. 1, pp. 113-130, 1986.
74. S. R. Bushell et al., "Wzi is an outer membrane lectin that underpins group 1 capsule assembly in *Escherichia coli*," *Structure*, vol. 21, no. 5, pp. 844-853, 2013.
75. S. A. Dehaghani et al., "An efficient purification method for high recovery of recombinant human granulocyte colony stimulating factor from recombinant *E. coli*," *Int J Environ Sci Dev*, vol. 1, no. 2, pp. 111-114, 2010.
76. R. Harrison (Ed.), *Protein purification process engineering* (Vol. 18), CRC Press, 1993.
77. H. G. Vikis and K. L. Guan, "Glutathione-S-transferase-fusion based assays for studying protein-protein interactions," *Protein-protein interactions: methods and applications*, pp. 175-186, 2004.
78. Lebediker, M. and Danieli, T., 2017. Purification of proteins fused to maltose-binding protein. In *Protein Chromatography: Methods and Protocols* (pp. 257-273).
79. S. Eshaghi et al., "An efficient strategy for high-throughput expression screening of recombinant integral membrane proteins," *Protein Science*, vol. 14, no. 3, pp. 676-683, 2005.
80. C. Chipot et al., "Perturbations of native membrane protein structure in alkyl phosphocholine detergents: a critical assessment of NMR and biophysical studies," *Chemical reviews*, vol. 118, no. 7, pp. 3559-3607, 2018.
81. N. A. Burgess-Brown et al., "Codon optimization can improve expression of human genes in *Escherichia coli*: a multi-gene study," *Protein Expression and Purification*, vol. 59, no. 1, pp. 94-102, 2008.

82. B. DeLaBarre et al., "Full-length human glutaminase in complex with an allosteric inhibitor," *Biochemistry*, vol. 50, no. 50, pp. 10764-10770, 2011.
83. Ayyar, B. V., Arora, S., Murphy, C., & O'Kennedy, R. (2012). Affinity chromatography as a tool for antibody purification. *Methods*, 56(2), 116-129.
84. Urh, M., Simpson, D., & Zhao, K. (2009). Affinity chromatography: general methods. *Methods in enzymology*, 463, 417-438.
85. W. Ott et al., "Single-molecule force spectroscopy on polyproteins and receptor–ligand complexes: The current toolbox," *Journal of structural biology*, vol. 197, no. 1, pp. 3-12, 2017.
86. R. Reese et al., "Novel peptide ligands for antibody purification provide superior clearance of host cell protein impurities," *Journal of Chromatography A*, vol. 1625, p. 461237, 2020.
87. Di Marco et al., "Simultaneous monitoring of monoclonal antibody variants by strong cation-exchange chromatography hyphenated to mass spectrometry to assess quality attributes of rituximab-based biotherapeutics," *International journal of molecular sciences*, vol. 22, no. 16, p. 9072, 2021.
88. D. Agyei et al., "Bioprocess challenges to the isolation and purification of bioactive peptides," *Food and Bioproducts Processing*, vol. 98, pp. 244-256, 2016.
89. D. L. Wyborski, J. C. Bauer, C.-F. Zheng, K. Felts, and P. Vaillancourt, "An *Escherichia coli* expression vector that allows recovery of proteins with native N-termini from purified calmodulin-binding peptide fusions," *Protein Expression and Purification*, vol. 16, no. 1, pp. 1-10, 1999.
90. J. Porath and B. Olin, "Immobilized metal ion affinity adsorption and immobilized metal ion affinity chromatography of biomaterials. Serum protein affinities for gel-immobilized iron and nickel ions," *Biochemistry*, vol. 29, pp. 1621-1630, 1983.
91. J. Hearon, "The configuration of cobalthistidine and oxy-bis(cobalthistidine)," *J. Natl. Cancer Inst.*, vol. 9, pp. 1-11, 1948.
92. M. N. Baeshen, A. M. Al-Hejin, R. S. Bora et al., "Production of biopharmaceuticals in *E. coli*: current scenario and future perspectives," *J Microbiol Biotechnol*, vol. 25, pp. 953-962, 2015.
93. B. Miroux and J. E. Walker, "Over-production of proteins in *Escherichia coli*: mutant hosts that allow synthesis of some membrane proteins and globular proteins at high levels," *J. Mol. Biol.*, vol. 260, pp. 289-298, 1996.
94. L. Dumon-Seignovert, G. Cariot, and L. Vuillard, "The toxicity of recombinant proteins in *Escherichia coli*: a comparison of overexpression in BL21(DE3), C41(DE3), and C43(DE3)," *Protein Expr. Purif.*, vol. 37, pp. 203-206, 2004.

95. J. Doherty, B. A. Connolly, and A. F. Worrall, "Overproduction of the toxic protein, bovine pancreatic DNaseI, in *Escherichia coli* using a tightly controlled T7-promoter-based vector," *Gene*, vol. 136, pp. 337-340, 1993.
96. H. Dong, L. Nilsson, and C. G. Kurland, "Gratuitous overexpression of genes in *Escherichia coli* leads to growth inhibition and ribosome destruction," *J. Bacteriol.*, vol. 177, pp. 1497-1504, 1995.
97. C. Gustafsson, S. Govindarajan, and J. Minshull, "Codon bias and heterologous protein expression," *Trends Biotechnol.*, vol. 22, pp. 346-353, 2004.
98. L. E. McDaniel and E. G. Bailey, "Effect of shaking speed and type of closure on shake flask cultures," *Appl. Microbiol.*, vol. 17, pp. 286-290, 1969.
99. F. J. Cui, Y. Li, Z. H. Xu, H. Y. Xu, K. Sun, and W. Y. Tao, "Optimization of the medium composition for production of mycelial biomass and exo-polymer by *Grifola frondosa* GF9801 using response surface methodology," *Bioresour. Technol.*, vol. 97, pp. 1209-1216, 2006. doi: 10.1016/j.biortech.2005.05.005
100. P. G. Blommel, K. J. Becker, P. Duvnjak, and B. G. Fox, "Enhanced bacterial protein expression during auto-induction obtained by alteration of lac repressor dosage and medium composition," *Biotechnol. Prog.*, vol. 23, pp. 585-598, 2007. doi: 10.1021/bp070011x
101. D. L. Hartley and J. F. Kane, "Properties of inclusion bodies from recombinant *Escherichia coli*," *Biochem. Soc. Trans.*, vol. 16, pp. 101-102, 1988.
102. M. M. Carrio and A. Villaverde, "Construction and deconstruction of bacterial inclusion bodies," *J. Biotechnol.*, vol. 96, pp. 3-12, 2002. doi: 10.1016/S0168-1656(02)00032-9
103. Derman, W. A. Prinz, D. Belin, and J. Beckwith, "Mutations that allow disulfide bond formation in the cytoplasm of *Escherichia coli*," *Science*, vol. 262, pp. 1744-1747, 1993. doi: 10.1126/science.8259521
104. Schein and M. Noteborn, "Formation of soluble recombinant proteins in *Escherichia coli* is favored by lower growth temperature," *Nat. Biotechnol.*, vol. 6, p. 3, 1988. doi: 10.1038/nbt0388-291
105. J. A. Vasina and F. Baneyx, "Expression of aggregation-prone recombinant proteins at low temperatures: a comparative study of the *Escherichia coli* cspA and tac promoter systems," *Protein Expr. Purif.*, vol. 9, pp. 211-218, 1997. doi: 10.1006/prep.1996.0678
106. Vera, N. Gonzalez-Montalban, A. Aris, and A. Villaverde, "The conformational quality of insoluble recombinant proteins is enhanced at low growth temperatures," *Biotechnol. Bioeng.*, vol. 96, pp. 1101-1106, 2007. doi: 10.1002/bit.21218

107. N. Gonzalez-Montalban, E. Garcia-Fruitos, and A. Villaverde, "Recombinant protein solubility - does more mean better?" *Nat. Biotechnol.*, vol. 25, pp. 718-720, 2007. doi: 10.1038/nbt0707-718
108. M. Martinez-Alonso, N. Gonzalez-Montalban, E. Garcia-Fruitos, and A. Villaverde, "The functional quality of soluble recombinant polypeptides produced in *Escherichia coli* is defined by a wide conformational spectrum," *Appl. Environ. Microbiol.*, vol. 74, pp. 7431-7433, 2008. doi: 10.1128/AEM.01446-08
109. H. Rosenberg, B. N. Lade, D. S. Chui et al., "Vectors for selective expression of cloned DNAs by T7 RNA polymerase," *Gene*, vol. 56, pp. 125-135, 1987.
110. E. Sulkowski, "Purification of proteins by IMAC," *Trends Biotechnol.*, vol. 3, pp. 1-7, 1985.
111. S. Knecht, D. Ricklin, A. N. Eberle, and B. Ernst, "Oligohis-tags: Mechanisms of binding to Ni²⁺-NTA surfaces," *J. Mol. Recognit.*, vol. 22, pp. 270-279, 2009.
112. T. Lanio, A. Jeltsch, and A. Pingoud, "Automated purification of His6-tagged proteins allows exhaustive screening of libraries generated by random mutagenesis," *BioTechniques*, vol. 29, pp. 338-342, 2000.
113. B. Muller-Hill, L. Crapo, and W. Gilbert, "Mutants that make more lac repressor," *Proc. Natl. Acad. Sci. U S A*, vol. 59, pp. 1259-1264, 1968.
114. S. E. Cellitti, D. H. Jones, L. Lagpacan, X. Hao, Q. Zhang, H. Hu, S. M. Brittain, A. Brinker, J. Caldwell, B. Bursulaya et al., "In vivo incorporation of unnatural amino acids to probe structure, dynamics, and ligand binding in a large protein by nuclear magnetic resonance spectroscopy," *J. Am. Chem. Soc.*, vol. 130, pp. 9268-9281, 2008.
115. S. A. Pless and C. A. Ahern, "Unnatural amino acids as probes of ligand-receptor interactions and their conformational consequences," *Annu. Rev. Pharmacol. Toxicol.*, vol. 53, pp. 211-229, 2013.
116. J. Siurkus, J. Panula-Perala, U. Horn, M. Kraft, R. Rimseliene, P. Neubauer, "Microbial Cell Fact.", vol. 9, p. 35, 2010.
117. Q. Long, X. Liu, Y. Yang, L. Li, L. Harvey, B. McNeil, Z. Bai, "J. Biotechnol.", vol. 192, pp. 323-338, 2014.
118. S. J. Wewetzer, M. Kunze, T. Ladner, B. Luchterhand, S. Roth, N. Rahmen, R. Klob, A. Costa e Silva, L. Regestein, J. Buchs, "J. Biol. Eng.", vol. 9, p. 9, 2015.
119. M. Levin, H. Otten, C. von Wachenfeldt, M. Ohlin, "BMC Biotechnol.", vol. 15, p. 52, 2015.
120. R. Khalilzadeh, S. A. Shojaosadati, N. Maghsoudi, J. Mohammadian Mosaabadi, M. R. Mohammadi, A. Bahrami, N. Maleksabet, M. A. Nassiri-Khalilli, M. Ebrahimi, E. H. Naderimanesh, "J. Ind. Microbiol. Biotechnol.", vol. 31, pp. 63-69, 2004.

121. S. Mazumdar, S. Sachdeva, V. S. Chauhan, S. S. Yazdani, "Bioprocess Biosys. Eng.", vol. 33, pp. 719-730, 2010.
122. S. Bhuvanesh, A. Chakkaravarthi, K. Perumal, R. Subramanian, "J. Ind. Microbiol. Biotechnol.", vol. 37, pp. 1053-1059, 2010.
123. N. K. Tripathi, A. Shrivastava, K. C. Biswal, P. V. Rao, "Appl. Microbiol. Biotechnol.", vol. 95, pp. 1179-1189, 2012.
124. Y. Wang, W. Ren, D. Gao, L. Wang, Y. Yang, Q. Bai, "Biomed. Chromatogr.", vol. 29, pp. 305-311, 2015.
125. C. Mateo, G. Fernandez-Lorente, B. C. Pessela, A. V. Carrascosa, J. L. Garcia, J. M. Guisan, "Affinity chromatography of polyhistidine tagged enzymes," J. Chromatogr. A, vol. 915, pp. 97-106, 2001. doi: 10.1016/s0021-9673(01)00626-4
126. F. Schäfer, N. Seip, B. Maertens, H. Block, J. Kubicek, "Purification of GST-Tagged Proteins," Methods Enzymol., pp. 127-139, 2015. doi: 10.1016/bs.mie.2014.11.005
127. M. Futatsumori-Sugai, R. Abe, M. Watanabe, M. Kudou, T. Yamamoto, D. Ejima, K. Tsumoto, "Utilization of Arg-elution method for FLAG-tag based chromatography," Protein Expr. Purif., vol. 67, pp. 148-155, 2009. doi: 10.1016/j.pep.2009.03.012
128. T. R. Butt, S. C. Edavettal, J. P. Hall, M. R. Mattern, "SUMO fusion technology for difficult-to-express proteins," Protein Expr. Purif., vol. 43, pp. 1-9, 2005. doi: 10.1016/j.pep.2005.03.016
129. U. Deuschle, W. Kammerer, R. Gentz et al., "Promoters of *Escherichia coli*: a hierarchy of in vivo strength indicates alternate structures," EMBO J., vol. 5, pp. 2987-2994, 1986.
130. C. A. Sommer, F. Henrique-Silva, M. T. M. Novo, "Teaching molecular biology to undergraduate biology students: An illustration of protein expression and purification," Biochem. Mol. Biol. Educ., vol. 32, pp. 7-10, 2004.
131. R. Shatzman, "Approaches to the expression of foreign genes," in Recombinant DNA Principles and Methodologies (J. J. Greene, V. B. Rao, eds.), Marcel Dekker, New York, pp. 551-578, 1998.
132. Soares-Costa, L. M. Beltramini, O. H. Thiemann, F. Henrique-Silva, "A sugarcane cystatin: Recombinant expression, purification, and antifungal activity," Biochem. Biophys. Res. Commun., vol. 296, pp. 1194-1199, 2002.
133. D. S. Hage, "Affinity chromatography: a review of clinical applications," Clin. Chem., vol. 45, pp. 593-615, 1999.

PROJECT DETAILS

Student Details			
Student Name	Manthan S Joshi		
Register Number	190924042	Section / Roll No	20
Email Address	manthan.joshi@learner.manipal.edu	Phone No (M)	+91 6366954555
Project Details			
Project Title	Enhancing Solubilization and Purification of M4: A Recombinant Membrane Protein Domain.		
Project Duration	Six months	Date of reporting	2 January 2023
Organization Details			
Organization Name	Tranalab Pvt Ltd, Centre for Human Genetics		
Full postal address with pin code	Biotech Park, Electronics City Phase 1, Electronic City, Bengaluru, Karnataka 560100		
Website address	http://www.chg.res.in/www.chg.res.in/index.html		
Supervisor Details			
Supervisor Name	Dr Vaishnavo Pai		
Designation	Lead Scientist		
Full contact address with pin code	Tranalab Pvt Ltd, Centre for Human Genetics, Biotech Park, Electronics City Phase 1, Electronic City, Bengaluru, Karnataka 560100		
Email address	vaishnavo.pai@tranalab.com	Phone No (M)	+91 9052217567
Internal Guide Details			
Faculty Name	Dr Subbalaxmi S		
Full contact address with pin code	Dept of Biotechnology, Manipal Institute of Technology, Manipal – 576 104 (Karnataka State), INDIA		
Email address	subbalaxmi.s@manipal.edu		

final

ORIGINALITY REPORT

15%

SIMILARITY INDEX

12%

INTERNET SOURCES

6%

PUBLICATIONS

4%

STUDENT PAPERS

MATCH ALL SOURCES (ONLY SELECTED SOURCE PRINTED)

Exclude quotes Off

Exclude bibliography On

Exclude matches < 10 words